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黑盒環境下聯邦學習的後門攻擊防禦

Defense against Backdoor Attacks for Federated Learning in Blackbox Setting

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摘要

聯邦學習使多個客户端能夠在不交換原始資料的情況下共同訓練全域模型，但仍然容易受到後門攻擊的威脅，攻擊者可在維持良好乾淨準確率的同時，悄然植入惡意行爲。由於許多現有的防護方法仰賴梯度、模型內部參數或中間資訊，而這些內容在嚴格的黑箱與隱私保護環境中通常無法取得，使得後門攻擊的偵測變得更加困難。本論文提出一種完全運作於僅觀測預測結果之黑箱環境的後門緩解方法，伺服器僅利用模型在小型且乾淨的驗證資料集上的輸出機率進行分析。該方法透過預測不確定性來刻畫各客户端的行爲特徵，並結合以中位數爲核心的強健統計量來辨識可能隱含後門行爲的異常預測模式，在假設多數客户端爲良性的前提下，仍能於對抗性環境中維持穩定性。本文進一步於多種代表性後門攻擊情境下進行評估，涵蓋靜態觸發式資料投毒、以訓練目標驅動的後門注入，以及不同攻擊設定之影響。實驗結果顯示，在符合實際應用的非獨立同分佈聯邦學習環境中，所提出的方法能在維持高乾淨準確率的同時，有效且穩定地降低攻擊成功率。整體而言，本研究證明基於預測層級不確定性的分析能作爲一種有效且實際可行的指標，用以提升嚴格黑箱限制下聯邦學習系統對後門攻擊的韌性。

關鍵詞：聯邦學習、後門攻擊、預測熵、強健統計、黑箱設定、適應型攻擊者、非同分佈。

ABSTRACT

Federated learning (FL) allows several clients to work together to train a global model without exchanging raw data, but it is still susceptible to backdoor attacks that can secretly introduce malevolent behavior while maintaining pristine accuracy. Because many current protections rely on gradients, logits, or internal model updates that are unavailable under stringent black-box limitations, detecting such assaults is especially difficult in privacy-preserving deployments. In this thesis, a backdoor mitigation approach is proposed that runs completely in a prediction-only black-box environment, with the server observing only model output probabilities on a clean, small validation set. The technique uses robust median-based statistics, namely the Median Absolute Deviation (MAD), to discover aberrant prediction patterns suggestive of backdoor activity after characterizing client behavior using Shannon entropy. By relying on strong statistics, the system maintains stability in adversarial conditions, assuming a benign majority of participating clients. Static trigger-based poisoning, optimization-driven backdoor injection, and varied attack settings are among the representative backdoor attack behaviors against which the suggested method is assessed. The approach reliably decreases attack success rates while maintaining strong clean accuracy, as shown by experimental results under realistic non-IID federated learning circumstances. The findings demonstrate that prediction-level uncertainty can serve as an effective and realistic indicator for backdoor resilience in federated learning systems working under strict black-box limitations.

Keywords: Federated learning, backdoor attacks, entropy, robust statistics, black-box setting, adaptive attacker, non-IID.

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Chapter 1

Introduction

1.1 Background

With the ongoing development of deep learning methods and computing resources, collaborative machine learning has gained significant importance in contemporary data-centric applications. Effective learning systems generally necessitate extensive and varied datasets to attain good generalization, especially in intricate tasks like image recognition and natural language processing. In real-world applications, though, such data is often scattered across various organizations or devices, each subject to privacy obligations, ownership limitations, and regulatory guidelines, rendering centralized data collection both impractical and hazardous.

Federated learning (FL) has been introduced as a decentralized training approach to tackle these issues by allowing several clients to collaboratively train a unified global model without sharing their raw data directly. In a typical FL framework, every client conducts local training on its confidential dataset and periodically sends model updates to a central server for aggregation, as originally outlined in communication-efficient federated optimization frameworks [1, 2]. By maintaining data localization, FL minimizes the direct exposure of sensitive data and has evolved as a highly regarded method

for privacy-conscious learning in fields with stringent data protection standards.

Although federated learning has its benefits, it does not automatically ensure security. The server's lack of direct access to data and training practices on the client-side means that adversaries can take advantage of this limited visibility to disrupt the learning process. Previous research has indicated that malicious participants can introduce harmful behavior during local training, which can affect the global model while remaining challenging to detect with standard aggregation methods.

Among the different risks associated with federated learning, backdoor attacks are particularly alarming due to their subtle and enduring characteristics. In these attacks, a malicious client incorporates concealed behaviors into the model by tainting local training data, leading the global model to function normally with harmless inputs while generating predictions controlled by the attacker when a specific trigger is activated. This dual functionality enables backdoor attacks to slip past traditional validation techniques, and recent studies have shown that these attacks can remain effective even with federated aggregation and limited defensive measures in place. [3, 4]

In practical federated learning systems, the risk of backdoor attacks is intensified by the variability of data among clients. When data is non-independent and non-identically distributed, it leads to differences in local model performance, complicating the identification of malicious alterations versus innocent variations. Although various Byzantine-robust aggregation techniques have been suggested to counteract poisoning attacks [2, 5, 6], their success frequently depends on having access to client updates and can suffer in effectiveness when faced with diverse data distributions.

Furthermore, numerous real-world deployments of federated learning function under stringent privacy regulations that restrict server-side visibility into gradients, model parameters, or loss values. In these rigorous black-box environments, the server is limited to observing prediction outputs produced by client models when evaluated on a small reference dataset [2, 7, 8], which considerably restricts the use of existing defense

strategies. These limitations highlight a significant disconnect between theoretical security measures and the practical requirements for deployment.

As a result, guaranteeing resilience against backdoor attacks under stringent black-box conditions has emerged as a significant and difficult research issue. Tackling this problem necessitates defensive strategies that depend exclusively on visible outputs while maintaining toughness against data diversity and agile adversaries. This thesis is placed within this framework and centers on creating a backdoor-resistant federated learning structure that functions entirely within strict black-box limitations without sacrificing privacy or model effectiveness.

1.2 Motivation

In recent times, federated learning has attracted considerable interest as a decentralized training approach that facilitates collaborative model development while maintaining data privacy and locality. By enabling clients to keep their raw data on-site and only share information related to the model, federated learning offers a feasible method for training machine learning models across various organizations that must adhere to strict privacy and regulatory standards. This approach is especially appealing in sensitive fields such as healthcare, finance, and personalized mobile applications. [1, 2]

However, despite these benefits, federated learning is not inherently immune to adversarial attacks. Among the numerous threats discussed in previous research, backdoor attacks represent a particularly serious danger due to their covert and enduring characteristics. In these types of attacks, malicious clients introduce hidden behaviors into the global model while maintaining high accuracy on legitimate inputs, which complicates detection through typical validation methods, especially challenging [3, 9]. As federated learning systems expand to real-world applications, the possible repercussions of undiscovered backdoors become increasingly alarming.

Current defense mechanisms against backdoor attacks in federated learning primarily focus on evaluating client updates, gradients, or the parameters of the internal model. While these approaches can improve resilience in controlled situations, they fundamentally rely on having access to internal data that is frequently unavailable in actual implementations due to privacy concerns. Moreover, their effectiveness diminishes in non-IID federated settings, where the behavior of benign clients naturally shows significant variability. These limitations considerably limit the practicality and implementation of numerous existing defenses in realistic federated learning environments.

Importantly, numerous practical federated learning applications function under stringent black-box constraints, where the server lacks access to gradients, model weights, or loss metrics and is only able to see prediction outputs. In such situations, conventional aggregation-based defenses are impractical. This discrepancy underscores the necessity for defense strategies that work exclusively with observable outputs, without compromising privacy protections or depending on reliable auxiliary components.

Driven by these issues, this thesis investigates a different approach to backdoor defense that focuses on analyzing prediction-level behavior. Rather than examining internal updates, the suggested method utilizes prediction uncertainty as a signal for detecting anomalous client behavior. Previous research indicates that models with backdoors frequently demonstrate unusual confidence patterns due to overfitting related to trigger features, which can be identified through output statistics.

In this scenario, strong statistical measures like the median and median absolute deviation offer a solid foundation for differentiating between benign and malicious clients, even when faced with adversarial influence and data variability. This leads to the creation of a streamlined, server-side defense solution that effectively mitigates backdoors while upholding the essential privacy standards of federated learning. Through this strategy, the thesis seeks to connect practical applicability with strong security within stringent black-box federated learning settings.

1.3 Contributions

This thesis outlines several key contributions stemming from the motivation behind this study.

1. **A Novel Black-Box Backdoor Detection Framework Tailored for Federated Learning:** This research introduces an innovative framework for detecting backdoors specifically crafted for strict black-box federated learning contexts, where the server does not have access to client-side gradients, model parameters, or loss values. By relying solely on prediction outputs produced by the involved clients on a limited clean validation set, the proposed method adheres to realistic constraints of privacy-preserving deployments. In contrast to existing defenses that operate under assumptions of partial model transparency, this framework functions entirely at the prediction level, rendering it apt for real-world federated learning systems that require stringent privacy measures.

2. **Entropy-Driven Client Behavior Profiling with Strong Statistical Filtering:** This thesis presents an entropy-driven method for profiling client prediction behavior in a strictly black-box environment. By measuring prediction uncertainty, this approach identifies behavioral anomalies linked to backdoor attacks. To differentiate unusual prediction behaviors from the normal fluctuations caused by non-IID data distributions, robust statistical metrics, particularly the median and median absolute deviation, are utilized. This framework allows for the consistent and dependable identification of suspicious clients without depending on heuristic thresholds, external trusted components, or internal model information, even when faced with adaptive adversaries and diverse client data.

3. **Assessment in Realistic Federated Learning Environments:** The proposed method's effectiveness is confirmed through comprehensive experiments carried

out in realistic federated learning contexts, which feature heterogeneous and non-IID data distributions, including instances of adaptive and persistent backdoor attacks. The results from these experiments indicate that the proposed framework significantly lowers attack success rates while maintaining high accuracy on clean data and overall model performance. These results underscore the robustness and practical usefulness of the suggested black-box defense in actual federated learning situations.

Chapter 2

Related Work

This chapter examines previous research related to backdoor attacks and defensive strategies within federated learning. It encapsulates notable attack methodologies, current defensive measures, and their shortcomings in practical implementation scenarios. The analysis sets the stage for integrating the proposed technique within the existing body of research.

2.1 Backdoor Attacks in Federated Learning

Backdoor attacks pose one of the most critical security risks in federated learning due to their covert nature and lasting consequences. In contrast to untargeted poisoning attacks that aim to impair overall model efficacy, backdoor attacks are crafted to instill specific malicious behaviors that remain inactive under standard conditions. When a specific trigger is present, the trained global model delivers outputs predetermined by the attacker while sustaining high performance on clean inputs, rendering such attacks challenging to identify during training and evaluation.

The feasibility of backdoor attacks in federated learning was first demonstrated by Bagdasaryan et al., who showed that even a small number of dishonest clients could

successfully establish a backdoor by amplifying their local updates to dominate the aggregation process [9]. Their study highlighted a significant vulnerability in standard federated averaging, where aggregation occurs without adequately validating client behavior. This finding pointed out that privacy-preserving training does not automatically guarantee protection against targeted model tampering [5].

Subsequent investigations broadened this threat model by examining more realistic and distributed attack strategies. Xie et al. proposed Distributed Backdoor Attacks (DBA), where several malicious clients work together to introduce partial triggers that collectively activate the backdoor [3]. This approach greatly enhances the stealth and persistence of the attack, particularly in scenarios where individual client updates are minimal. Such coordinated assaults make detection more complex [10], as no single harmful update appears significantly out of the ordinary.

Moving beyond fixed poisoning strategies, later studies emphasized the importance of persistence and adaptability. Neurotoxin introduced a long-lasting backdoor attack that selectively suppresses benign neuron updates while enhancing features linked to the backdoor [4]. This design allows the backdoor to survive through numerous federated rounds, even if malicious clients participate sporadically. Recent research has further indicated that backdoor behaviors can continue to exist even under federated optimization schemes intended to stabilize training dynamics, suggesting that such attacks are not easily thwarted by conventional algorithmic modifications [11]. These results imply that backdoors in federated learning can retain their effectiveness long after the initial poisoning incident.

Recent investigations have also demonstrated that backdoor attacks can be carried out under progressively stricter conditions. CapsuleBD indicated that attackers can successfully implant backdoors in heterogeneous federated learning environments, where clients use a variety of model architectures [12]. This finding challenges the belief that architectural diversity inherently improves security. Concurrently, black-box attack sce-

narios are gaining traction, these approaches assume stronger server-side observability, a scenario that mirrors real-world deployments more closely and significantly undermines the effectiveness of gradient-based defenses [13, 14]. These attacks provide a more accurate representation of actual operational conditions and considerably reduce the efficacy of defenses based on gradient analysis.

By suggesting more subtle, flexible, and persistent attack techniques, recent research has greatly improved our knowledge of backdoor attack potential in federated learning. In order to show that backdoor behaviors may be flexibly incorporated without depending on predefined trigger semantics, 3DFed presents an extensible framework for creating covert backdoors that adjust to various training and deployment settings [15]. Effective backdoor injection is still possible in realistic federated circumstances, such as partial client involvement and non-IID data distributions, according to practical backdoor attack evaluations [16]. In addition to static triggers, attackers can increase attack reliability while decreasing detectability across clients by using coordinated and model-dependent trigger designs [17]. The limits of static mitigation techniques are highlighted by adaptive attack schemes like A3FL, which explicitly adapt poisoning behavior to the training dynamics and deployed protections [18]. More recent research has examined collusive and covert poisoning tactics, in which several compromised clients work together to maintain backdoor effectiveness with enhanced concealment [19], as well as irreversible backdoor objectives that endure across rounds and withstand mitigation efforts [20]. Collectively, these studies show that contemporary backdoor assaults in federated learning provide significant obstacles for strong defensive design since they are becoming more adaptable, persistent, and challenging to identify.

The sophistication of attacks has further progressed with the introduction of frequency-domain triggers and optimization-driven strategies. Low-frequency backdoor attacks embed triggers into spectral components that are difficult to detect, while LADDER frames backdoor injection as a multi-objective optimization problem that balances at-

tack effectiveness, stealth, and durability [21]. These advancements underscore a clear movement towards more adaptive and evasive backdoor strategies. This thesis examines three typical attack contexts that exhibit different adversarial behaviours and fit our experimental design among the many backdoor tactics investigated in federated learning.

Recent studies indicate that backdoor attacks in federated learning are not only achievable but also becoming more robust in realistic deployment situations, such as non-IID data distributions, limited client participation, varying model architectures, and stringent black-box conditions [10]. This changing threat environment highlights the importance of defense strategies that do not rely on strong visibility assumptions and can effectively counter a wide range of flexible attack methods.

In addition to particular image-based backdoors, recent studies have explored backdoor attacks that exploit the semantic and distributional characteristics of the training data. Label replacement attacks modify class labels during local training to create consistent misclassification without requiring obvious visual triggers, rendering them particularly challenging to detect [22]. Because the contaminated samples appear visually indistinguishable from benign data, it becomes especially difficult to separate malicious behavior from natural client variability in federated learning.

Another significant area of research examines backdoor attacks under conditions of limited or no information. DarkFed has shown that attackers are capable of embedding effective backdoors without having access to authentic training data by generating substitute inputs during local optimization [23]. This greatly reduces the hurdles for adversaries and demonstrates that simply limiting data access does not suffice to avert backdoor implantation. These findings underscore that backdoor attacks can be carried out even when attackers possess constrained or incomplete knowledge about the global data distribution.

More recently, studies have focused on the relationship between backdoor attacks and the dynamics of federated learning. Adaptive attackers can take advantage of mech-

anisms for client selection, schedules for learning rates, and the frequency of aggregation to enhance attack persistence while reducing the likelihood of detection. By aligning their updates with benign statistical trends, such attackers can bypass robust aggregation rules and exert sustained influence over the global model [24]. These attackers can bypass robust aggregation rules and exert sustained influence over the global model. These insights indicate that effective defenses need to consider not just static attack characteristics but also the strategic adaptation that occurs over federated iterations.

2.2 Defense Strategies in Federated Learning

This study focuses on a black-box federated learning setup that is strictly prediction-only, meaning that the server does not have access to internal optimization states, gradients, model updates, client-side training data, or logits. Prediction probability outputs from server-side assessment on a limited, locally stored reference dataset that is not shared with clients make up the sole visible data. Stronger presumptions that are not applicable in this context include gradient access, trusted validation sharing, and protocol alterations, which are the foundation of many backdoor protections now in use.

As the susceptibility of federated learning systems to backdoor attacks became more apparent, a significant amount of research arose focusing on defensive tactics aimed at reducing harmful client actions during collaborative training. Unlike centralized learning, defenses in federated learning must function under stringent privacy requirements, restricted visibility, and diverse client data distributions. These traits fundamentally influence both the development and efficacy of existing defense strategies.

Initial defense efforts primarily focused on the aggregation stage of federated learning, positing that malicious actions manifest as unusual or extreme client updates. Robust aggregation methods were among the first proposed defenses aimed at counteracting poisoning attacks. Blanchard et al. introduced Krum, which selects a subset of

client updates that are nearest to the majority within parameter space, thereby excluding outlier updates before aggregation [25, 26].

Despite Krum offering theoretical guarantees for robustness against Byzantine failures, later research has demonstrated that its basic assumptions frequently fail in practical federated learning scenarios. Specifically, in the case of non-IID data distributions, legitimate client updates may inherently diverge, leading Krum to mistakenly classify valid updates as outliers [27]. Furthermore, recent evaluations suggest that aggregation techniques based on geometric proximity have difficulty differentiating malicious updates from innocent statistical variations in decentralized contexts, which restricts their effectiveness in real-world implementations [5].

To improve tolerance to outliers, an alternative method known as trimmed mean aggregation was introduced, which discards extreme values prior to averaging. Although this technique bolsters resistance to basic poisoning attacks [28], later studies revealed that well-crafted backdoor updates could still remain within permissible statistical ranges, enabling malicious clients to evade detection [24]. Additionally, both Krum and trimmed mean necessitate direct approaches that assume stronger server-side observability, which restricts their applicability in privacy-sensitive contexts [7]. In addition to combining statistics, scholars have explored defenses that utilize supplementary signals or reliable information. FLTrust implemented a small server-stored trustworthy dataset to calculate a reference update, which is subsequently used to assess the reliability of client contributions during aggregation [29]. By anchoring aggregation choices to an established benign signal, this method enhances resilience against backdoor attacks and other forms of poisoning. [5]. However, depending on trusted data brings additional assumptions and does not negate the necessity of directly examining client updates.

Validation-based defenses further refine this concept by evaluating the impact of client updates on clean validation data. BAFFLE screens questionable updates based on

their effect on validation performance, demonstrating improved detection accuracy in controlled conditions [30]. Despite its efficiency, this approach raises computational requirements and assumes access to intermediate model states, which might not be present in real federated learning scenarios with stringent privacy protections [7].

Recent research has also looked into adaptive and temporal defenses that monitor client activity over training rounds. ADFL utilizes historical update trends to detect suspicious behaviors that signify poisoning, thereby enhancing defense against ongoing backdoor attacks [31]. Nonetheless, these temporal defenses are still susceptible to adaptive adversaries who intentionally modify their update behavior to imitate benign clients, allowing them to avoid detection over time [24]. Furthermore, these strategies continue to depend on direct access to client updates, highlighting the ongoing disparity between current defense mechanisms and the limitations of practical federated learning settings.

In summary, existing defense mechanisms have considerably improved the robustness of federated learning against backdoor attacks. However, their dependence on internal model information, trusted auxiliary elements, or strong statistical assumptions constrains their effectiveness in practical circumstances. These shortcomings prompt a more thorough investigation of defense strategies capable of functioning under tighter observability constraints.

The availability of internal model information at the server is a fundamental premise shared by the majority of security tactics, notwithstanding their diversity. Specifically, to calculate similarity metrics, identify outliers, or assess update consistency, most aggregation-based defenses operate under assumptions that differ from the strict prediction-only black-box setting considered in this work. Although these presumptions might be true in controlled experimental settings, privacy laws, encrypted computation, or hardware-enforced isolation frequently violate them in real-world federated learning implementations.

These methods operate under assumptions that differ from the strict prediction-only black-box setting considered in this work, only a small subset of previous defenses can be deemed applicable. Many techniques nevertheless rely implicitly on auxiliary trusted components or intermediary optimization states, even when they claim to have lower observability requirements. This gap drives additional research into prediction-level and behavior-based approaches by highlighting the necessity for protection strategies that can function dependably without examining client updates. The behavioral goal of current backdoor assaults is to enforce confidence misclassification toward attacker-specified labels while maintaining clean accuracy, despite differences in poisoning techniques, trigger designs, and optimization tactics. Instead of attack-specific signatures, this common goal appears as systematic distortions in prediction uncertainty. As a result, under stringent black-box limitations, uncertainty-based indicators like prediction entropy provide a behavioral abstraction that is less susceptible to the implementation of a backdoor, making them appropriate for assessing robustness across a variety of attack tactics.

2.2.1 Defenses within the Limits of Privacy and Observability Constraints

In real-world federated learning implementations, the server often faces significant limitations in monitoring client behavior. Privacy laws, encrypted communication methods, and proprietary client devices can restrict access to stronger server-side observability. In these situations, many established defense mechanisms become impractical since they fundamentally depend on analyzing internal client-side data.

Numerous investigations have highlighted the gap between theoretical defense models and real-world application scenarios. In particular, assessments of federated learning security indicate that many defenses based on aggregation unnaturally rely on white-box access to client updates—an assumption that does not hold in systems focused on pri-

vacy [32]. Even with limited access, scrutinizing gradients or intermediate states raises concerns regarding potential privacy infringements, adherence to legal requirements, and the preservation of user trust.

These complexities are further exacerbated in heterogeneous federated learning environments. Non-IID data distributions lead to discrepancies in typical client behavior, making it difficult to distinguish between harmful updates and legitimate ones. As a result, robust aggregation techniques may unintentionally impair the performance of non-malicious clients, adversely influencing overall model effectiveness, especially in situations with heterogeneous data. At the same time, adaptive attackers can exploit this variability to conceal their actions. Empirical studies have shown that adaptive poisoning strategies can evade static or threshold-based defenses by incrementally adjusting attack intensity or selectively engaging in specific training rounds to align with benign update patterns [24].

To address these issues, recent research has begun to explore defense mechanisms that function under reduced assumptions regarding server-side observability. Notably, Sun et al. introduced the notion of strict black-box federated learning, where the server can only access prediction outputs generated by client models on a limited reference dataset [7]. This scenario closely resembles realistic implementations in which increasing observability might jeopardize privacy safeguards. Under such restrictions, conventional aggregation- and update-based defenses lose their applicability.

Interestingly, Dong et al. presented a black-box backdoor detection method with minimal data and information. Their design similarly uses robust dispersion measures (such median/MAD) and entropy statistics to distinguish anomalous behaviors [33]. In a more recent study, Miao et al. examined backdoor resistance in privacy-preserving federated learning environments where the server might not have direct access to model parameters, inspiring countermeasures that hold up even when server-side visibility is limited [34]. These studies show that mitigating black-box backdoors is possible, but

they also emphasize the necessity of precisely defining what signals are viewable and how aggregation is carried out under stringent limitations.

While the strict black-box framework is pertinent to real-world systems, it is still not well-explored in the context of backdoor mitigation. Current methods that assert black-box suitability often still depend on auxiliary probing techniques, structured input transformations, or additional communication overload, which may not be practical [8]. More generally, thorough evaluations of federated learning security have pointed out that most existing defenses presuppose a greater level of observability than is available in realistic deployments, creating a significant gap for backdoor-resilient mechanisms that operate solely at the prediction level while remaining robust against data variability and adaptive adversaries [2].

To clarify the practical assumptions that underpin current aggregation-based defenses, a table provides a detailed comparison of representative methods in federated learning. The table highlights the extent of server-side observability that each method requires, including access to client updates, reliance on trusted auxiliary information, and the necessity of reference data. It also indicates whether a method can function within strict black-box federated learning conditions. Unlike this work, these approaches assume stronger server-side observability. Recent evaluations have underscored that many federated learning defenses implicitly assume access to stronger server-side observability, which restricts their use in privacy-preserving environments [32]. More recent studies oriented towards black-box frameworks have reinforced this limitation by demonstrating that aggregation and verification strategies that depend on inspecting updates or maintaining parameter-space consistency fail when the server is limited to feedback at the prediction level [35]. This comparison illustrates that most existing defenses operate under white-box or grey-box conditions, highlighting the necessity for aggregation methods that remain efficient under conditions of prediction-only observability.

The aforementioned problems show a basic mismatch between realistic federated learning environments and current backdoor defense techniques from the standpoint of practical deployment. Many defenses work well under white-box or grey-box assumptions, but when the server is limited to prediction-only observability, they become useless. Defenses based on gradient inspection, parameter-space consistency, or update-level validation cannot be implemented in such environments without going against privacy requirements.

This thesis explicitly focuses on rigorous black-box federated learning. These methods operate under assumptions that differ from the strict prediction-only black-box setting considered in this work in order to address this unresolved constraint. The suggested method uses prediction-level uncertainty patterns to characterize client behavior instead of trying to deduce malicious behavior from inaccessible internal signals. The suggested approach, which combines robust statistical filtering with entropy-based behavioral profiling, is made to function well in situations where many current defenses deteriorate or fail, such as minimal observability, non-IID data distributions, and adaptive adversarial techniques.

2.2.2 Recent SOTA Backdoor Defense Frameworks in Federated Learning

Recent research has suggested specific backdoor mitigation methods that specifically target harmful client behavior in federated learning, going beyond traditional Byzantine-robust aggregation restrictions. Without depending on gradient inspection, FLShield presents a validation-based protection architecture that uses server-side evaluation signals to detect tainted client updates [36]. In order to retain privacy guarantees and visible signals for mitigation, SafeLearning focuses on safe aggregation with improved backdoor detectability [37]. Multi-stage or synergistic mitigation methods that include screening, aggregation control, and collaborative decision mechanisms are the focus of

more modern collaborative defense frameworks like FL-CDF and FLSDA. These tactics decrease backdoor success while maintaining clean model utility [38, 39]. Furthermore, to increase resilience against backdoor injection, defense strategies based on bidirectional elections and individual client viewpoints provide organized interaction mechanisms among participants [40]. Even while these techniques show promise, many of them depend on extra presumptions like greater observability, protocol tweaks, or trusted validation access, which would not work in rigorous prediction-only black-box federated learning environments.

Name	Category	Server Observability	Client Updates	Validation Set	Server-Side Reference Dataset	Behavioral Signal Used	Black-Box Compatible
FedAvg [1]	No Defense	Full model updates	✓	×	×	None	×
krum [25]	Classical Robust	Full model updates	✓	×	×	Update distance consistency	×
Fltrust [29]	Grey-box	Updates + trusted reference	✓	✓	×	Reference update similarity	×
RFout [41]	White-box SOTA	Full model updates	✓	×	×	Gradient-based outlier statistics	×
TAG [42]	Recent SOTA	Updates + Reference	✓	×	✓	Reference-induced update impact	×
Proposed	**Black-box**	Predictions only	×	×	✓	**Prediction entropy**	✓

2.3 Constraints in Black-Box Environments

While current defense mechanisms have shown some effectiveness against backdoor attacks under specific conditions, their usefulness is greatly diminished when federated learning operates under stringent black-box constraints. In numerous real-world applications, the rigorous prediction-only black-box setup taken into consideration in this study is not the same as the assumptions made by these approaches, or loss values because of privacy laws, encrypted communication, or hardware restrictions. In these scenarios, a significant portion of the existing research loses its practicality.

Defenses that rely on aggregation fundamentally hinge on the direct evaluation of client updates. Methods such as robust and dynamic aggregation functions explicitly necessitate access to gradients or parameter updates to compute similarity metrics, mitigate outliers, or adjust aggregation rules based on observed update behaviors [43]. In the absence of gradient access, these defenses cannot be employed, no matter how theoretically valid they are. Moreover, even validation-based approaches, which utilize clean reference data to assess client reliability, typically require access to intermediate model states or the impacts of updates, implicitly violating strict black-box conditions.

Client behavior modeling techniques encounter similar limitations. Defenses that rely on temporal or clustering-based methods analyze historical update patterns to identify anomalies; however, these techniques presume access to client updates across multiple rounds. In strictly black-box environments, these approaches assume stronger server-side observability, rendering these methods unfeasible. Furthermore, adaptive adversaries can alter their attack strategies to resemble benign clients, enabling them to evade detection even when visibility is limited [24, 28].

An additional challenge emerges from the data heterogeneity present in federated learning. In the case of non-IID data distributions, benign clients inherently demonstrate varied training dynamics and prediction behaviors. This inconsistency makes it difficult to distinguish between malicious and benign actions, leading to an increase in false

positive rates or diminished model performance for update-based defenses. Empirical research suggests that the effectiveness of aggregation-based methods tends to decline as data heterogeneity rises [6], especially when these defenses are developed or adjusted based on IID assumptions.

Recent research has introduced the notion of stringent black-box federated learning, in which the server can exclusively see prediction outputs generated by client models on a restricted reference dataset. While Miao et al. stress that privacy-preserving FL can further limit server access to model information, necessitating defenses that remain effective without direct parameter observability [34], Dong et al. show that prediction-level uncertainty statistics can be used to detect backdoor behaviors under limited information [33]. This situation reflects realistic deployment conditions but underscores a significant shortcoming in existing defenses [44]. Although various attacks have been shown to be effective in black-box settings, particularly adaptive and heterogeneous backdoor attacks [45], relevant defense mechanisms remain largely unexamined.

Adaptive adversaries may still try to imitate benign prediction uncertainty on the server-side reference dataset, which drives thorough screening and thoughtful threat-model discussion in subsequent chapters.

In conclusion, available backdoor defenses either depend on access to sensitive internal information, impose extra trust assumptions, or do not generalize effectively under realistic non-IID and adaptive adversarial situations. The lack of reliable defenses under strict black-box conditions underscores a fundamental weakness in present federated learning security frameworks. Tackling this issue necessitates a transition from update-centric evaluations to defense mechanisms that rely solely on observable model behavior, which inspires the prediction-based approach discussed in this thesis.

Chapter 3

Preliminary

This chapter outlines the essential concepts and system assumptions that underpin this research, including federated learning, mechanisms for backdoor attacks, and constraints related to black-box observability. These introductory elements create the theoretical and operational backdrop for the defense framework that will be elaborated upon in the following chapters.

3.1 Federated Learning

Federated Learning (FL) represents a distributed machine learning approach that allows a number of clients to work together to train a common model without the need to share their individual datasets directly. This framework is fundamental to numerous current federated learning systems and forms the basis for later aggregation and resilience strategies aimed at managing diverse data distributions, untrustworthy clients, and malicious activities [28,43,46]. Rather than consolidating data, each client conducts training locally and transmits only model updates to a coordinating server, making FL particularly advantageous in environments where privacy and regulatory compliance are critical.

Federated Learning Configuration

Consider a federated learning framework involving N clients represented as

$$\mathcal{C} = \{C_1, C_2, \ldots, C_N\},$$

where each client C_i possesses a confidential local dataset D_i with a size of $n_i = |D_i|$. Let $n = \sum_{i=1}^{N} n_i$ represent the total number of training samples across all clients. The goal of federated learning is to collaboratively develop a global model characterized by $\mathbf{w}$ that minimizes the weighted empirical risk across all clients:

$$\min_{\mathbf{w}} \; F(\mathbf{w}) = \sum_{i=1}^{N} \frac{n_i}{n} F_i(\mathbf{w}), \tag{3.1}$$

where

$$F_i(\mathbf{w}) = \mathbb{E}_{(\mathbf{x},y)\sim D_i}\left[\ell(f_{\mathbf{w}}(\mathbf{x}), y)\right]$$

represents the local loss function for client i, $f_{\mathbf{w}}(\cdot)$ is the model's prediction mechanism, and $\ell(\cdot)$ denotes a loss specific to the task.

Federated Averaging (FedAvg)

The most commonly used optimization algorithm in federated learning is *Federated Averaging (FedAvg)*. Training occurs in synchronous communication rounds indexed by $t = 1, 2, \ldots$, following a server-client interaction model.

During round t, the server manages the global model $\mathbf{w}^{(t)}$ and follows these steps:

1. **Global Model Distribution.** The server shares the current global model $\mathbf{w}^{(t)}$ with a selected group of participating clients.

2. **Local Optimization.** Each chosen client initializes its local model with the global parameters received and conducts local training on its private data. The local

update (e.g., using stochastic gradient descent) is calculated as

$$\mathbf{g}_i^{(t)} = \nabla F_i(\mathbf{w}^{(t)}), \tag{3.2}$$

where $\mathbf{g}_i^{(t)}$ signifies the gradient computed using client i's local dataset.

3. **Aggregation at Server.** The server compiles the received updates from clients by using a weighted average based on the sizes of their datasets:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \sum_{i=1}^{N} \frac{n_i}{n} \mathbf{g}_i^{(t)}, \tag{3.3}$$

with η representing the learning rate.

4. **Synchronization of Model.** The refreshed global model $\mathbf{w}^{(t+1)}$ is redistributed to clients for the upcoming communication round.

Algorithm 1 outlines the typical Federated Averaging (FedAvg) process utilized as the foundational optimization technique in federated learning. This decentralized training approach maintains data locality and mitigates privacy concerns related to sharing raw data. However, it also presents new security issues, as the server depends on client updates that cannot be completely verified for their integrity and intentions. Previous research has pointed out that this vulnerability allows for adversarial actions such as model poisoning and backdoor attacks, especially when a select group of clients acts maliciously.

Algorithm 1 Federated Averaging (FedAvg)

Require: Number of clients K; client fraction $C \in (0, 1]$; local minibatch size B; local epochs E; learning rate η; total rounds T

Ensure: Global model after T rounds: $w^{(T)}$

```
Server: initialize global model w^(0)
for t = 1, 2, ..., T do
    m ← max(⌊CK⌋, 1)
    S_t ← randomly sampled set of m clients
    for all client k ∈ S_t in parallel do
        w_k^(t) ← ClientUpdate(k, w^(t), B, E, η)
    end for
    w^(t+1) ← Σ_{k∈S_t} (n_k / Σ_{j∈S_t} n_j) w_k^(t)
end for
return w^(T)
procedure ClientUpdate(k, w, B, E, η)
    Split local dataset P_k into minibatches of size B
    for e = 1, 2, ..., E do
        for all minibatch b ∈ P_k do
            w ← w − η∇ℓ(w; b)
        end for
    end for
    return w
end procedure
```

3.2 Formulation of Backdoor Attacks in Federated Learning

Backdoor attacks represent a specific category of poisoning attacks that occur at training time, where an attacker intends to incorporate hidden harmful behaviors into a model without compromising its performance on untainted inputs [9]. The key trait of a backdoor attack is that its malicious behavior is triggered solely by a particular pattern, rendering these attacks especially subtle and challenging to uncover.

Within the realm of federated learning, backdoor attacks pose an even greater danger due to the distributed nature of the training process and the absence of direct oversight over client-side data and optimization methods [3]. Because the server compiles updates from various clients without evaluating their semantic meanings, malevolent

actions taken by a small group of participants may be integrated into the global model without any immediate decline in accuracy on clean data.

In a standard scenario involving federated backdoor attacks, the adversary has control over a subset of the clients participating, denoted as $\mathcal{A} \subseteq \{C_1, \ldots, C_N\}$. Each malicious client aims to introduce a concealed backdoor into the global model by altering its local training procedures while maintaining satisfactory performance on benign data. A prevalent tactic is to insert a trigger pattern into certain local samples and change their labels to an adversary-selected target class. This method of trigger-based poisoning is commonly utilized in federated learning backdoor attacks because of its effectiveness and discretion.

Let $\mathcal{T}(\cdot)$ represent the trigger transformation function, and let $y^\star$ denote the target label specified by the attacker. For a malicious client $C_i \in \mathcal{A}$ with a local dataset D_i, the contaminated dataset can be formally specified as

$$D_i^{\text{poison}} = \{(\mathcal{T}(\mathbf{x}), y^\star) \mid (\mathbf{x}, y) \in D_i\}, \tag{3.4}$$

where the clean inputs are altered by the trigger function and re-labeled to enforce the backdoor action.

The effective training dataset utilized by the malicious client subsequently becomes a combination of clean and poisoned samples, described by

$$D_i' = D_i \cup D_i^{\text{poison}}. \tag{3.5}$$

By training on D_i', the malicious client optimizes for both standard classification accuracy on clean inputs and the attacker-defined response on triggered inputs, which facilitates the gradual embedding of the backdoor into the global model during the aggregation process.

During the local training phase, malicious clients adjust their model parameters to

concurrently achieve performance on both clean classification tasks and backdoor activation goals. This dual process can be expressed as a composite optimization problem:

$$\min_{\mathbf{w}_i} \; \mathcal{L}_{\text{clean}}(\mathbf{w}_i; D_i) + \lambda \, \mathcal{L}_{\text{bd}}(\mathbf{w}_i; D_i^{\text{poison}}), \tag{3.6}$$

where $\mathcal{L}_{\text{clean}}$ represents the standard task loss, $\mathcal{L}_{\text{bd}}$ encourages misclassification of triggered inputs to $y^\star$, and λ determines the intensity of the backdoor objective [4].

After completing local optimization, each client sends its trained model to the server. Let round $\mathbf{w}^{(t)}$ represent the global model at time round t, and $\mathbf{w}_i^{(t,E)}$ indicate the model trained locally after E epochs. The client update can be expressed as

$$\Delta \mathbf{w}_i^{(t)} = \mathbf{w}_i^{(t,E)} - \mathbf{w}^{(t)}. \tag{3.7}$$

Since traditional aggregation methods like FedAvg treat all client updates equally, poisoned updates from malicious clients can impact the global model and progressively embed the backdoor through successive rounds of communication."

Features of the attack: Federated backdoor attacks possess several traits that set them apart from traditional poisoning attacks. Firstly, they are *discreet*, as the overall accuracy of the global model is mostly unchanged. Secondly, they are *enduring*, with backdoor behaviors continuing to exist even when malicious clients participate sporadically. Third, they exhibit *versatility*, enabling adversaries to modify poisoning rates, trigger configurations, or update magnitudes in accordance with normal statistical patterns, thereby evading resilient aggregation methods [24].

These attributes render federated backdoor attacks particularly difficult to counter, especially in realistic scenarios where the server has limited insights into client-side information. This dilemma underscores the necessity for defense strategies that function under limited or black-box observation conditions, which will be elaborated upon in the following sections.

3.3 Prediction Behavior in Neural Networks

The predictive behavior of neural networks describes how a trained model associates an input sample with a distribution over output classes. In classification tasks, contemporary neural networks usually generate a vector of real-valued scores, commonly known as logits, which represent the model's preference for each class. These logits are then converted into a probabilistic form for better interpretation and decision-making.

For a given input sample $\mathbf{x}$ and a model defined by $\mathbf{w}$, the predicted class probabilities are derived using the softmax function, which normalizes the logits to create a valid probability distribution:

$$p(y = k \mid \mathbf{x}; \mathbf{w}) = \frac{\exp(z_k(\mathbf{x}))}{\sum_{j=1}^{C} \exp(z_j(\mathbf{x}))}, \tag{3.8}$$

where $z_k(\mathbf{x})$ indicates the logit for class k, and C represents the total number of classes.

In addition to the predicted label, the shape of the output probability distribution offers crucial insights into the model's confidence and level of uncertainty. Shannon entropy is a widely used quantitative metric for assessing prediction uncertainty [47], as it captures the degree of concentration or dispersion of the probability mass across classes. For a specific input $\mathbf{x}$, the entropy of the prediction distribution is expressed as:

$$H(\mathbf{x}) = -\sum_{k=1}^{C} p(y = k \mid \mathbf{x}) \log p(y = k \mid \mathbf{x}). \tag{3.9}$$

Low entropy values signify confident predictions that are largely dominated by a single class, while higher entropy values indicate ambiguity or uncertainty among several classes.

Under favorable training conditions, neural networks generally demonstrate stable and consistent prediction behavior on clean validation data, with entropy values displaying a relatively smooth and predictable distribution. However, adversarial training

scenarios can severely disrupt this behavior. In particular, backdoor attacks modify the learned decision boundaries to maintain high accuracy on clean inputs while enforcing behavior dictated by the attacker on triggered samples. This manipulation can result in abnormal confidence patterns, which may include excessively sharp or inconsistent probability distributions when tested on selected inputs.

Crucially, prediction behavior can be analyzed without needing access to the model's internal parameters, gradients, or training updates. By querying a trained model using a fixed, server-controlled validation set and gathering only the output probabilities, it's feasible to examine behavioral traits in a purely black-box manner [8]. This observation model aligns well with practical federated learning implementations, where privacy, security, or system restrictions prevent inspection of client-side updates.

Backdoor attacks lead to unique prediction signatures. While clean samples continue to produce confident and accurate predictions to retain utility, triggered inputs often show a highly concentrated probability mass on the target class specified by the attacker. Additionally, poisoned client contributions may cause subtle instability or increased uncertainty in predictions over clean reference data, especially during the initial training rounds [48]. These anomalies are not always discernible from accuracy metrics alone but can be uncovered through distributional analysis of prediction outputs.

These findings underscore the importance of uncertainty-aware statistics, such as entropy-based measures, to characterize and differentiate between benign and malicious client behaviors. Unlike inspection techniques that rely on gradients or parameters, prediction-level analysis depends solely on observable outputs, thus remaining compatible with strict black-box federated learning assumptions. This perspective forms the foundation for the prediction-centric monitoring approach outlined later in this thesis.

Normal (Benign) Prediction Behavior. Under benign training settings, neural networks display robust and reliable prediction behavior on clean validation data. With greater entropy values for ambiguous or borderline samples and lower entropy values

for well-separated cases, the projected probability distributions typically reflect the intrinsic complexity of the input. Crucially, as long as the training goals stay in line with the desired task, this uncertainty behavior is comparatively smooth among clients even under non-IID data distributions.

Abnormal (Malicious or Backdoored) Prediction Behavior. Backdoor attacks, on the other hand, purposefully alter the learned decision boundaries in order to impose a particular attacker-defined response in the presence of a trigger pattern. For triggered inputs, when the probability mass becomes strongly concentrated on the target class, this modification frequently leads to overconfident predictions. Abnormally low entropy values that diverge from normal benign uncertainty patterns are indicative of such conduct. Even when malicious clients keep normal accuracy on clean data, these altered confidence characteristics produce detectable behavioral abnormalities in the model's prediction distributions.

3.4 Statistical Instruments for Analyzing Behavior

3.4.1 Entropy as a Measure of Uncertainty

Prediction uncertainty offers crucial insights into how confidently a model assigns class labels to input samples. A sharply peaked prediction distribution signifies a high level of confidence, while a flatter distribution suggests greater uncertainty. Shannon entropy is a commonly used measure for quantifying this uncertainty and has proven effective in detecting anomalous prediction behavior in learning systems.

For a specified input sample $\mathbf{x}$, let the predicted class probability vector be $\mathbf{p}(\mathbf{x}) = (p_1(\mathbf{x}), \ldots, p_C(\mathbf{x}))$, where C represents the total number of classes. The Shannon en-

tropy for this prediction is defined as:

$$H(\mathbf{p}(\mathbf{x})) = -\sum_{k=1}^{C} p_k(\mathbf{x}) \log p_k(\mathbf{x}), \tag{3.10}$$

which measures the uncertainty in the model's prediction for the input $\mathbf{x}$. In practice, a small constant $\epsilon > 0$ is incorporated during implementation to ensure numerical stability when calculating the logarithm.

In the context of federated learning, the server assesses client models using a fixed validation set $V = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ that is owned by the server. Let $\mathbf{p}_i(\mathbf{x})$ represent the prediction distribution generated by client i's model for input $\mathbf{x} \in V$. The average prediction entropy for client i is then calculated as:

$$\bar{H}_i = \frac{1}{m} \sum_{\mathbf{x} \in V} H(\mathbf{p}_i(\mathbf{x})). \tag{3.11}$$

This average entropy score reflects the overall uncertainty behavior of client i across the clean reference inputs and serves as the foundation for further robust client analysis.

From an intuitive perspective, entropy offers a succinct and instructive synopsis of how strongly a model reacts to incoming samples. Strong model confidence is shown by a low entropy when the projected probability distribution is heavily concentrated on one class. On the other hand, entropy rises when probability mass is dispersed throughout several classes, indicating ambiguity or uncertainty in the model's choice. Such uncertainty patterns, even in the presence of non-IID data distributions, are statistically consistent among clients and naturally correlate to the semantic complexity of the input data under benign training conditions.

This uncertain behavior is radically changed by backdoor attacks. Malicious training specifically builds a strong link between the trigger pattern and an attacker-specified target label to ensure dependable backdoor activation. This procedure suppresses alter-

native class probabilities and results in abnormally low entropy values by encouraging the model to make very confident predictions when the trigger is present. Because of this, even when their accuracy on clean data is essentially unchanged, backdoored models show unique confidence profiles that differ from the benign population [9, 49].

Importantly, entropy-based characterization does not rely on access to internal optimization states such as gradients, parameters, or update magnitudes. Instead, it operates solely on observable prediction probabilities, making it particularly well-suited for strict black-box federated learning environments. Prior studies have demonstrated that entropy is an effective indicator of anomalous or adversarial model behavior, especially in settings involving distribution shifts, confidence miscalibration, or adversarial manipulation [8, 48]. Compared to accuracy- or loss-based metrics, entropy captures the full distributional structure of predictions, enabling the detection of subtle behavioral deviations that are otherwise difficult to observe.

3.4.2 Robust Statistics

In adversarial federated learning, a subset of clients might act maliciously, generating entropy scores that significantly diverge from the majority. Traditional statistical metrics like the mean and variance are quite sensitive to such outliers. In contrast, robust statistics offer reliable estimates even when a portion of the data is compromised.

For a collection of average entropy scores $\{\bar{H}_i\}_{i\in\mathcal{S}^{(t)}}$ from the clients participating in communication round t, the median entropy is defined as:

$$\mathrm{med}^{(t)} = \mathrm{median}\left(\{\bar{H}_i\}_{i\in\mathcal{S}^{(t)}}\right). \tag{3.12}$$

The Median Absolute Deviation (MAD), which acts as a reliable measure of spread, is calculated as:

$$\mathrm{MAD}^{(t)} = \mathrm{median}\left(\left\{\left|\bar{H}_i - \mathrm{med}^{(t)}\right|\right\}_{i\in\mathcal{S}^{(t)}}\right). \tag{3.13}$$

Using these statistics, clients whose entropy scores differ significantly from the population are identified through the following thresholding criterion:

$$\left|\bar{H}_i - \mathrm{med}^{(t)}\right| > \alpha \cdot \mathrm{MAD}^{(t)}, \tag{3.14}$$

where $\alpha > 0$ determines the strictness of the filtering guideline.

Clients that meet this criterion are classified as abnormal and are excluded from the subsequent aggregation processes. Due to its resilience against outliers, this MAD-based filtering approach remains effective even when some clients exhibit malicious behavior. When applied to entropy scores obtained solely from prediction outputs, it enables a consistent analysis of client behavior without the need to access model parameters, gradients, or internal states, thus maintaining the strict black-box framework.

3.5 Black-Box Observation Framework

3.5.1 Strict Black-Box Presumption

This thesis is based on a *strict black-box observation framework* concerning federated learning. Within this model, the server is not allowed to access any internal client information, such as model parameters, gradients, logits, intermediate activations, or the states of internal training. The server's interactions with the clients are limited to querying their trained models and observing the prediction probability outputs generated on a clean validation dataset that the server controls.

This constraint mirrors real-world federated learning implementations where privacy laws, secure environments, or encrypted computing protocols hinder direct access to client-side updates. As a result, the server cannot intervene in or modify the training procedures of clients and must depend solely on observable behavior from the models externally.

Given the strict black-box assumption, the federated learning system is subject to the following restrictions:

- **Client updates are not visible:** The server has no access to gradients, model parameters, weight updates, or internal activations from local training.
- **No dependence on trusted reference models:** The server does not presume the existence of trustworthy clients or reference updates for validation.
- **Interaction confined to prediction outputs:** The server can query client models using a clean validation dataset it controls and can only observe the corresponding probability vectors produced.
- **Client training is unaltered:** Client-side training methods remain the same, ensuring participant independence and confidentiality of data.

Formally, let

$$V = \{x_j\}_{j=1}^{|V|}$$

represent a clean validation dataset held by the server. For each client i, the server only observes the resulting prediction probability outputs

$$\mathbf{p}_i(V) = \{\mathbf{p}_i(x_j) \mid x_j \in V\},$$

where $\mathbf{p}_i(x_j)$ indicates the softmax probability vector generated by the locally trained model of client i for the input x_j. The server cannot access any information beyond these prediction outputs.

This strict observation model guarantees that all subsequent analyses and aggregation strategies are compatible with privacy-preserving environments for federated learning and do not depend on unrealistic assumptions of access.

3.5.2 Challenges in Black-Box Federated Learning

In contrast to white-box or grey-box federated learning scenarios—where gradient vectors, update norms, or interior model statistics can be observed—the strict black-box framework significantly limits the server's capacity to assess client behavior. Many current methods for robust aggregation and backdoor defense fundamentally rely on such internal information, making them inapplicable under strict black-box conditions.

This limited observability leads to several challenges.

Limited visibility of malicious activity. Backdoor attacks are often crafted to remain inert on clean data and only activate under certain trigger conditions. Since the server only observes predictions on a clean validation dataset, malicious clients may seem indistinguishable from legitimate ones during regular inference activities.

Incompatibility of gradient-based defenses. A significant number of existing defenses depend on identifying anomalies in gradients, update sizes, or statistics at the layer level. In a strict black-box context, these indicators are completely unattainable, preventing the use of gradient-based or representation-level analyses.

Adaptive tactics from attackers. Malicious clients might intentionally adjust their prediction behavior on clean inputs to imitate benign clients, thus diminishing the effectiveness of simple thresholding or rule-based detection due to limited visibility.

Trade-off between robustness and utility. The server must manage the balance between robustness and model utility based only on behavioral patterns from prediction outputs. Being overly aggressive in filtering could lead to the exclusion of benign clients and reduced performance, while overly cautious strategies may fail to mitigate the influence of backdoors. Finding a balance under strict black-box constraints remains a key challenge.

These limitations prompt a *prediction-centric behavioral analysis* approach, where client actions are inferred solely from patterns of output uncertainty rather than from internal model conditions. Accordingly, this thesis utilizes prediction entropy and robust

statistical methodologies to support backdoor-resilient aggregation within the strict confines of black-box federated learning.

Chapter 4

System Architecture

This chapter explores the specifics of both the system and adversary models. By defining the architectural assumptions and threat boundaries, this chapter lays the groundwork for the proposed black-box defense framework.

4.1 System Model

Under rigorous black-box constraints, our experimental setup, shown in Fig. 4.1, conforms to the standard federated learning architecture. Many dispersed clients and a central federated server that manages the training process make up the architecture. While the server keeps a tiny auxiliary reference dataset in secret that is never shared with clients and is only used for server-side assessment, each client retains its own private local dataset.

1. Clients. Clients act as distributed entities that possess private local datasets, indicated as D_i. These datasets are never shared with the server or other clients. During each round of communication, selected clients conduct local model training using their private data and subsequently update their local models. Given the implemented strict black-box configuration, the server does not have access to any client-side gradients,

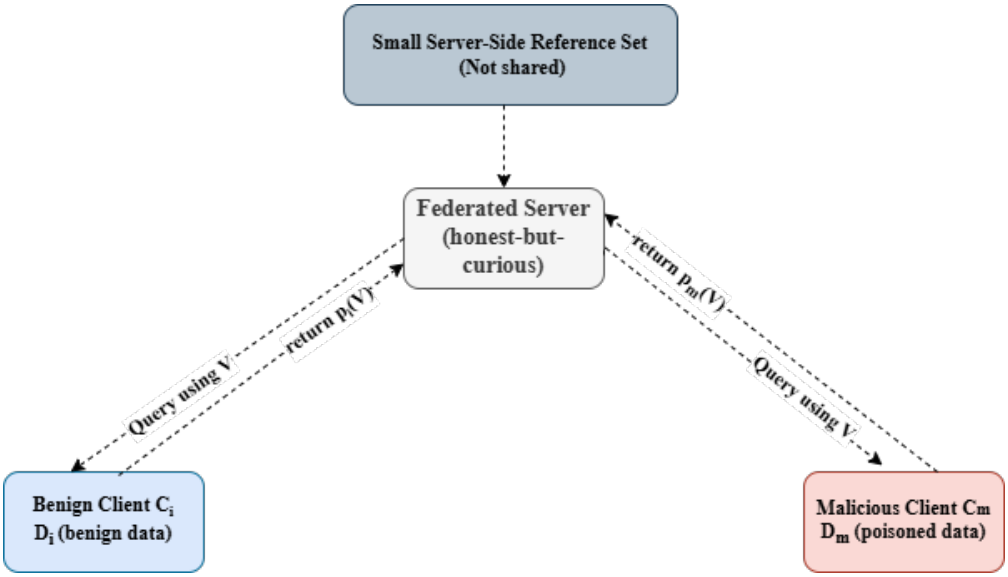

Figure 4.1: System model of federated learning under a strict black-box setting

model parameters, optimizer statuses, or internal activations.

Rather, participating clients upload their locally trained models or the accompanying prediction outputs to the server in a tight black-box environment. No server-held data is received, accessed, or assessed by clients. The server side is where all evaluation-related tasks are carried out. The prediction probability outputs produced by the client models are the sole information that is visible from the server's point of view. Clients can be malevolent or benign, but at the protocol level, the server cannot tell them apart.

2. Server. The central server manages the federated learning operation and is presumed to be *honest-but-curious*, a frequently adopted threat model in the federated learning domain.

Honest: Aggregation. The server diligently adheres to the federated learning protocol by coordinating the training rounds, distributing the global model to the selected clients, gathering their responses, and aggregating client contributions to refresh the global model. In this research, aggregation is executed using entropy-aware weighting after filtering out anomalous clients, which allows for robust global model updates while

maintaining learning utility. The refreshed global model is then redistributed to clients for the following training round.

Curious: Analysis. Although it strictly observes the protocol, the server may undertake an analysis of the observable data it receives. Such analysis is limited in this work to client identities and the client prediction probability outputs $\mathbf{p}_i(V)$ produced by client models when assessed by the server on its locally stored reference dataset. The server does not have access to client datasets D_i, gradients, model parameters, logits, or optimizer statuses. This limitation enforces a stringent black-box observability framework, under which all proposed analysis and mitigation measures within this thesis operate solely on prediction-level information.

The training process is repeated iteratively. The server uses its locally stored reference dataset to perform robust aggregation and entropy-based screening after distributing the current global model to a chosen group of clients and receiving their uploaded models or prediction results. This reference dataset does not take part in training or coordination; it is used only for screening and aggregation control. Strong privacy and security assurances are maintained while the system architecture appropriately represents genuine federated learning deployments by limiting server observability to prediction-level information only.

4.2 Adversary Model

This study examines a federated learning context characterized by stringent black-box observability, where the adversary acts from within the system rather than as an outsider. In line with previous research, the adversarial activity is represented by compromised clients that seek to introduce backdoors into the globally trained model while conforming to the normal federated learning process.

1. Adversary Capability: The adversary operates through one or more malicious

clients that comply with the federated learning protocol while injecting corrupt local data. These malicious clients have complete control over their local training data and objectives. Specifically, the adversary can taint its local datasets by incorporating backdoor triggers and assigning labels chosen by the attacker. During local training, these clients optimize their models to achieve both high accuracy on legitimate inputs and elevated success rates for attacks when the trigger pattern is activated.

Adversarial clients behave in the same way as valid clients in terms of protocol conformance, notwithstanding their evil purpose. As needed by the protocol, they upload their locally trained models or related prediction outputs after taking part in each communication round, receiving the global model from the server, and performing local training. No server-held reference data is sent to, accessed by, or evaluated by malicious clients. Importantly, they don't provide modifications to variables for the model, or internally trained states, therefore based only on protocol-level behavior, they can't be distinguished from benign clients.

The adversary is presumed to be adaptive, allowing for adjustments in poisoning strategies throughout the training rounds. However, it lacks insight into the server's defense thresholds or filtering decisions, and the adversary does not have access to or control over the server's privately kept reference dataset, nor does it know the server's defensive levels or filtering criteria.

2. Adversary Objective: The primary aim of the adversary is to establish a lasting backdoor in the global model, ensuring that the model functions normally with clean inputs while generating outputs predetermined by the attacker when a specific trigger is activated. This must be accomplished without causing significant degradation in the global model's clean accuracy, as noticeable declines in performance could lead to suspicion or diminish the effectiveness of the attack.

In addition to achieving a high rate of attack success, the adversary also seeks to avoid detection by server-side defense mechanisms. In the circumstances of this

research, this means that malicious clients must simulate benign prediction behavior when the server evaluates their uploaded models or prediction outputs on its locally stored reference dataset, all the while preserving the embedded backdoor's functionality. This dual goal creates an inherent challenge for the adversary, as excessively confident predictions caused by backdoor overfitting could result in abnormal prediction entropy, raising the likelihood of detection.

By outlining the adversary model in this way, the proposed defense is assessed against realistic and adaptable backdoor attacks within strict black-box parameters. The assumptions made in this context are in line with practical implementations of federated learning, where the server has limited visibility and malicious actions must be inferred solely from the model answers or prediction outputs that customers upload and have assessed on the server.

Chapter 5

Proposed Methodology

This chapter outlines the method put forward in this research. We introduce a black-box framework aimed at mitigating backdoors in federated learning, which functions under stringent observability limitations and depends exclusively on the prediction results from clients. The suggested approach aims to detect and diminish harmful contributions from clients while maintaining the effectiveness of legitimate updates. The complete process takes place on the server side and is segmented into a series of clearly defined stages.

5.1 Client Screening Based on Entropy

In this section, we present the client screening process based on entropy used by BLACKBOXGUARD within a strictly black-box federated learning framework. A tiny auxiliary reference dataset V is privately maintained by the server and is never made available to clients. In order to produce prediction probability vectors $\mathbf{p}_i(x)$ for each $x \in V$, participating clients upload their locally trained models or matching prediction outputs during each communication cycle. These are assessed only at the server. No client data D_i, gradients, model parameters, logits, or optimizer states are accessed, which aligns

with the stringent black-box federated learning assumptions where the server can solely view the prediction outputs [7]. The aim of this phase is to create a concise uncertainty signature for every client and to perform thorough statistical analysis to pinpoint clients displaying unusual prediction confidence behaviors that may be caused by backdoor attacks.

Prediction probability outputs in this study are presumed to be obtained via authentic model inference carried out on the server. Predictions are derived by server-side evaluation (or trusted inference); we do not presume that malevolent clients may arbitrarily create probability vectors independent of the uploaded model behavior. This presumption aligns with popular black-box federated learning environments that use trusted inference or server-side assessment.

Algorithm 2 outlines the full screening process. The server initially calculates the mean Shannon entropy for each client across the reference set, yielding a scalar entropy score H_i. Subsequently, it determines the median m and the Median Absolute Deviation (MAD) of the entropy scores, only keeping clients whose deviation from the median is within an α-scaled MAD threshold. The result of this screening phase is the set of indices for retained clients, denoted as $\mathcal{C}_{\text{keep}}$, which is then used in the entropy-aware aggregation phase.

Because entropy directly represents prediction confidence behavior, which is known to be altered by backdoor attacks, it is especially appropriate for client screening in this context. To ensure dependable backdoor activation, malevolent clients must enforce extremely confident predictions on triggered inputs, as covered in Chapter 3. This goal invariably modifies the model's uncertainty characteristics, resulting in entropy patterns that diverge from benign behavior. Entropy allows for efficient behavioral screening without having access to gradients, parameters, or internal training states by condensing prediction uncertainty into a single scalar number per client.

It is crucial to remember that no particular backdoor structure is assumed by the

suggested entropy-based screening method. It makes use of a common behavioral result of backdoor training: the establishment of anomalous prediction confidence, rather than trigger visibility or poisoning patterns. Successful backdoor attacks must impose confident misclassification toward an attacker-defined target, regardless of whether the attack is carried out by architecture-dependent modifications, adaptive optimization techniques, or static trigger injection. Since the prediction entropy reflects this confidence distortion, it is possible to identify it without explicitly understanding the attack technique.

It should be noted that prediction entropy is not used to determine per-sample correctness or to detect poisoned samples. Entropy, on the other hand, is employed at the client level as a behavioral indicator that captures systematic confidence patterns brought on by backdoor aims. Because of this, a malicious client may have aberrant entropy characteristics throughout the reference set while still achieving high accuracy on benign inputs.

5.1.1 Rationale for Median-Based Robust Filtering

The procedure for client screening based on entropy utilizes robust statistical methods to differentiate between normal client behavior and unusual patterns caused by backdoor attacks. This strategy is based on the understanding that malicious clients typically display significantly different prediction uncertainty on the server-side reference dataset, which appears as irregular entropy values.

Let us $\{H_i\}_{i=1}^{K}$ represent the entropy scores calculated for the clients participating in a specific communication round. Instead of calculating the mean and variance, which are very sensitive to outliers, the screening method uses the median as a stable measure of central tendency. The spread of entropy scores is assessed using the Median Absolute Deviation (MAD), which is defined as the median of the absolute differences from the median entropy. If the MAD is equal to zero, it indicates that all clients are retained

Algorithm 2 Entropy-Based Client Screening (Mean Entropy + Median-MAD)

Require: Server-side reference dataset V (not shared); client prediction probability vectors $\{\mathbf{p}_i(x)\}_{i=1}^{K}$ for all $x \in V$; threshold α

Ensure: Retained client index set $\mathcal{C}_{\text{keep}}$

Compute entropy scores
for $i = 1$ **to** K **do**
    $H_i \leftarrow \frac{1}{|V|} \sum_{x \in V} \left(- \sum_c p_{i,c}(x) \log p_{i,c}(x) \right)$
end for
Compute robust statistics
$m \leftarrow \text{median}(\{H_i\}_{i=1}^{K})$
$\text{MAD} \leftarrow \text{median}(\{|H_i - m|\}_{i=1}^{K})$
Screen clients
$\mathcal{C}_{\text{keep}} \leftarrow \emptyset$
for $i = 1$ **to** K **do**
    if $|H_i - m| \leq \alpha \cdot \text{MAD}$ **then**
        $\mathcal{C}_{\text{keep}} \leftarrow \mathcal{C}_{\text{keep}} \cup \{i\}$
    end if
end for
return $\mathcal{C}_{\text{keep}}$

since there is no detectable deviation in entropy.

Clients whose deviation in entropy from the median surpasses a threshold scaled by α are considered anomalous and are thus excluded from the aggregation process. This filtering approach functions entirely at the prediction level and does not necessitate access to the client's data, gradients, or model parameters, thereby maintaining strict adherence to the black-box model assumption. By implementing this robust screening approach before aggregation, the impact of malicious clients is diminished while the contributions of legitimate participants are preserved.

There are two different ways that entropy deviations might occur. Backdoor behavior that imposes confident trigger–target mappings is frequently linked to excessively confident and steeply peaked prediction distributions, which are correlated with abnormally low entropy. Atypically high entropy, on the other hand, indicates erratic or inconsistent prediction behavior, which might be the consequence of hostile attempts

to avoid discovery, poisoned optimization, or untrustworthy local training. Under the benign-majority assumption, both of these deviations are considered legitimate signs for client screening since they deviate from benign client behavior, even if their underlying reasons differ.

To maintain an effective backdoor, overconfident predictions about a target label must be persistently enforced throughout training rounds, even when an adaptive attacker may momentarily modify its behavior to resemble benign clients. Because they reduce the impact of fleeting or purposefully concealed actions while maintaining stability under the benign-majority assumption, median-based robust statistics are therefore ideally suited to capture such persistent aberrations.

5.1.2 Entropy as a Behavioral Signal under Black-Box Constraints

Strict black-box FL prevents the server from seeing user data or training components and only allows it to monitor prediction outputs. Such data is either unavailable because of system and security limitations or purposefully hidden to protect data privacy. As a result, many current defenses that depend on representation-level analysis, parameter distance measures, or gradient inspection are inapplicable in this context.

On the other hand, when assessed by the server using a reference dataset that is locally stored, prediction entropy may be calculated only from output probability distributions produced by client models. Entropy may be quantified without changing the federated learning protocol or compromising client privacy because prediction probabilities are naturally generated during inference. Therefore, under rigorous black-box limitations, entropy is one of the few behavioral signals associated to uncertainty that may still be seen [47, 50].

Entropy represents the confidence distribution of a model's predictions from a behavioral standpoint. Previous research has demonstrated that many backdoor attacks specifically seek to retain high accuracy on clean data while enforcing extremely confi-

dent trigger–target misclassifications [7, 16, 20]. In malevolent training, this behavior is usually accomplished by suppressing alternative class probabilities, which leads to abnormally low entropy values and artificially peaked output distributions when triggered inputs are supplied. Crucially, even in cases when clean data performance seems to be unaffected, such confidence distortions could still be detectable at the server level.

Therefore, rather than providing conclusive evidence of malicious conduct, prediction entropy functions as a probabilistic behavioural indicator that catches aberrations in output confidence patterns. Since entropy describes the distributional structure of prediction outputs, it is more susceptible to confidence manipulation brought on by backdoor training than accuracy-based metrics, which frequently hold steady despite backdoor attacks. Previous work has shown that backdoor attacks often introduce abnormal behavioral patterns in model predictions or feature distributions, which can be used to analyze poisoned models without accessing internal gradients or parameters [51].

Yet, indications based on entropy are not always trustworthy. Additionally, data heterogeneity, adaptive adversaries that specifically aim to avoid detection techniques based on confidence, or inherently challenging samples can all result in elevated or decreased entropy [52]. As a result, the suggested approach employs aberrant entropy as a screening signal rather than a stand-alone assurance of maliciousness and functions under typical federated learning assumptions, such as a benign-majority setting and a clean and representative server-side reference dataset. We specifically present the benign drop rate (false-positive rate) in the experimental assessment to further evaluate the dependability of entropy-based screening. This statistic quantifies the percentage of benign clients that the suggested screening process mistakenly filters. Furthermore, a median absolute deviation (MAD)-based rule that controls the robustness of the entropy threshold offers a statistically sound method for detecting anomalous entropy values under varied client behaviour. To make sure the filtering process is stable under various conditions, sensitivity to the threshold parameter is also examined. These assessments

show that the suggested screening method functions dependably under the benign majority assumption while elucidating the practical constraints of entropy-based signals. To further confirm the validity of the entropy-based screening mechanism, Chapter 6 presents an empirical examination of benign drop rate and entropy threshold behavior.

Ultimately, the suggested method provides a theoretically supported and empirically observable signal for aggregation control and client screening while naturally adhering to stringent black-box federated learning constraints by solely depending on prediction entropy calculated from output probabilities.

5.2 Entropy-Weighted Aggregation

In order to increase resilience under stringent black-box limitations, the suggested aggregation technique is built on the conventional Federated Averaging (FedAvg) architecture and enhanced with entropy-based client filtering and entropy-guided weighting.

Once the server has identified and removed anomalous clients using prediction entropy metrics, it proceeds to combine the contributions from the remaining clients with consideration for entropy. The goal of this phase is to reduce the impact of unreliable or questionable client behaviors while still acknowledging the input from trustworthy clients, all while adhering to strict black-box observability requirements. Notably, this aggregation approach solely depends on the prediction uncertainty calculated from the server-side reference dataset V (not shared) and does not involve accessing client-side gradients, model parameters, or optimizer states.

Prediction entropy shows how probability mass is spread across classes, which indicates confidence manipulation that accuracy alone cannot capture, in contrast to accuracy-based indicators, which frequently stay unaffected despite backdoor assaults. This distinction encourages the use of entropy as a weighting signal during aggregation as well as for client screening.

5.2.1 Weight Assignment Based on Entropy

The weighting step works inside the filtered set, $\mathcal{C}_{\text{keep}}$ as weights are only given to customers who are still there following MAD-based filtering. Let us $\mathcal{C}_{\text{keep}}$ represent the collection of clients that remain after the entropy-based filtering process. For each client $i \in \mathcal{C}_{\text{keep}}$, the server allocates an aggregation weight determined by the prediction entropy H_i of the client. Clients with lower entropy on the clean reference set are considered more reliable, as their predictions indicate greater confidence and alignment with expected benign behavior.

The weight that incorporates entropy is defined as follows:

$$w_i = \frac{H_{\max} - H_i}{H_{\max} - H_{\min} + \varepsilon}, \tag{5.1}$$

where $H_{\max} = \max_{j \in \mathcal{C}_{\text{keep}}} H_j$, $H_{\min} = \min_{j \in \mathcal{C}_{\text{keep}}} H_j$, and a little constant called ε is added to guarantee numeric stability. The above equation gradually reduces the influence of clients whose prediction uncertainty approaches the screening threshold while giving higher weights to clients with lower entropy.

5.2.2 Weighted Model Aggregation

By employing the weights derived from entropy, denoted as $\{w_i\}_{i \in \mathcal{C}_{\text{keep}}}$, throughout the aggregation process, the server adjusts the global model by modifying the contributions of participating clients. In a conceptual framework, the global model update during the communication cycle $t + 1$ can be expressed as

$$\theta^{(t+1)} = \frac{\sum_{i \in \mathcal{C}_{\text{keep}}} w_i \, \theta_i^{(t)}}{\sum_{i \in \mathcal{C}_{\text{keep}}} w_i}, \tag{5.2}$$

where $\theta_i^{(t)}$ indicates the model trained locally by client i. This equation outlines the logic of aggregation despite needing knowledge of inner optimization stages, gradients,

logits, or client training data, thus adhering to the stringent black-box assumption.

5.2.3 Robustness and Black-Box Compliance

The suggested aggregation technique uses entropy-based behavioral analysis to improve resilience while maintaining the fundamental structure of Federated Averaging. In particular, the collection of participating clients and their proportional contributions are modified based on prediction entropy rather than gradient- or parameter-level data, but the aggregation is still a weighted averaging of client models, in line with FedAvg.

Strict black-box federated learning assumptions, which limit the server's observability toThe server uses its locally stored reference dataset to assess prediction results, are fully met by this approach. Without examining client-side optimization specifics or internal model states, all aggregate decisions—including client filtering and contribution weighting—are solely based on observable prediction probabilities. Consequently, the suggested solution is suitable for actual federated deployments in which access to internal client information is controlled.

The aggregation technique successfully lessens the impact of malicious clients trying to introduce backdoors while maintaining the contributions of benign users by combining entropy-based filtering with entropy-guided weighting. Crucially, the convergence behavior and scalability characteristics of conventional federated learning are preserved since the aggregation mechanism is still FedAvg-based. One of the main design goals of the suggested technique is striking a compromise between robustness, practicality, and black-box compatibility.

The efficacy of the suggested approach is independent of presumptions on the attacker's internal strategy since it solely relies on prediction outputs. To guarantee the success of an attack, the induced prediction behavior must be consistent with the backdoor objective, even if adversaries modify their training objectives across rounds or take advantage of varied client architectures. The suggested aggregation technique preserves

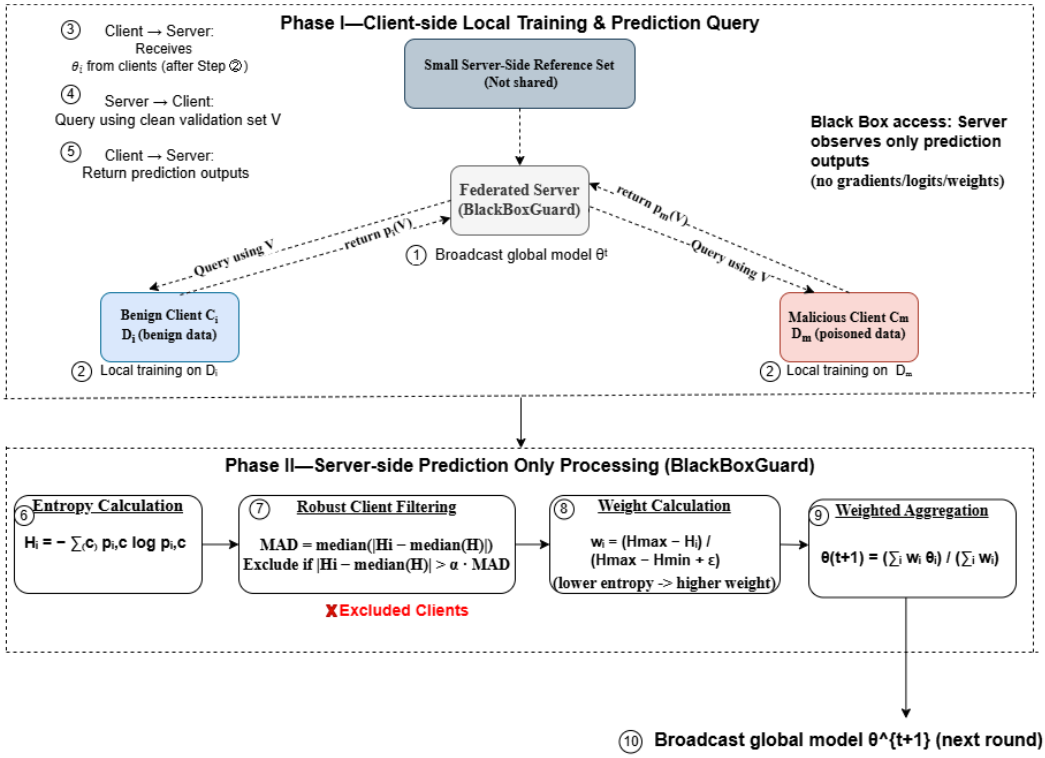

Figure 5.1: Overview of the entropy-based prediction filtering framework

strong black-box compliance while maintaining robustness across various backdoor behaviors by concentrating on entropy statistics rather than internal changes.

We observe that in a rigorous black-box situation, identifying such behavior would become intrinsically challenging if an attacker could precisely match the entropy distribution of benign clients across several rounds. All defenses that do not have access to gradients, model updates, or internal training states have this fundamental constraint of prediction-only observability.

Chapter 6

Experiments and Comparisons

6.1 Experimental Setup

6.1.1 Setup and Datasets

A central server manages a number of dispersed clients in all trials, which adhere to a common cross-device federated learning paradigm. The server broadcasts a global model to participating clients throughout each communication round. Depending on the defensive mechanism being assessed, clients train locally on their own data and deliver updated models or prediction results. After that, the server compiles client input to update the global model for the following cycle.

CIFAR-10 Setup (Main Benchmark): considering its greater visual complexity and extensive use in federated backdoor investigations, CIFAR-10 serves as the main benchmark for comparison with baseline protection techniques. Ten classes and 60,000 color pictures make up the dataset; 10,000 samples were utilized for testing and 50,000 for training. We take into consideration a small-scale federated environment with $K = 4$ clients for CIFAR-10 tests. In order to enable controlled testing under various malicious participation ratios (R30 and R50), a small-scale design is purposefully employed. This

configuration preserves repeatability and a reasonable processing cost while allowing accurate monitoring of attack-defence interactions. A disjoint subset of 3,000 training samples is randomly selected from the CIFAR-10 training set for each client. Every communication cycle involves participation from every client. Using stochastic gradient descent with momentum 0.9 and weight decay 10^{-4}, each client completes three local epochs throughout the 20 global rounds of training. The number of training rounds is selected to provide adequate model convergence while maintaining a reasonable experiment duration for repeated assessments across various assault and response strategies. For every trainable layer, the learning rate is set at 10^{-3}. For repeatability, a fixed random seed is employed in every experiment. A ResNet-18 backbone that has been trained with ImageNet is used to initialise the global model. Only the last convolutional block (`layer4`) and the classification head are adjusted, leaving the other layers fixed in order to stabilise training and lower communication overhead. To conform to the pre-trained model design, input photos are scaled 224×224 and normalised using ImageNet statistics.

MNIST Setup (Ablation and Behavioural Analysis): Ablation studies and detailed behavioural studies of the suggested defence, including computational overhead and transitory benign filtering, are conducted using MNIST. Seventy thousand grayscale handwritten digit pictures from ten classes make up the collection. A bigger federated configuration with 50 clients is used for MNIST tests. In order to more accurately simulate cross-device federated learning scenarios in which several dispersed players contribute to the global model, this bigger client population was chosen. Ten clients are chosen at random to take part in each round. Each chosen client uses a lightweight convolutional neural network to carry out two local training epochs every round during the 200 global communication rounds of training. To enable in-depth investigation of round-level behavioural patterns, such as entropy-based client filtering dynamics and benign client involvement over time, MNIST employs a longer training cycle. Stable

per-round analysis and more lucid observation of entropy-based client behaviour under controlled settings are made possible by this configuration. Malicious clients, if they exist, are corrected during training, and a benign-majority assumption is enforced in all settings. At the end of each cycle, the server aggregates client updates using defence-specific aggregation techniques or conventional federated averaging (FedAvg). While attack success rate (ASR) is tested using triggered test samples built in accordance with the associated attack configuration, which is explained in the next chapter, clean accuracy (ACC) is assessed on the standard test set.

6.1.2 Attack Model and Configurations

We examine many sample adversarial scenarios in federated learning to assess the resilience of the suggested approach and baseline defenses. Malicious clients change their local training process to incorporate backdoor behaviors while retaining high accuracy on benign inputs in all assaults, which are carried out under a data poisoning threat model. Malicious clients are fixed during training and take part in each communication round, unless otherwise indicated.

Small-scale federated setups are used to instantiate the malicious participation settings for CIFAR-10 studies. The R50 option uses four total clients with two malicious clients (2/4 = 50%), whereas the R30 setting uses three total clients with one malicious client (1/3 = 33%). For consistency between tests, we use the symbol R30 even though it equates to an exact ratio of 33.3% because of discrete client numbers. Every communication cycle involves participation from all CIFAR-10 clients. By inserting a fixed bottom-right trigger and relabeling those samples to the target class (label 0), each malicious client poisons 20% of its local data. In order to maintain the attack's effectiveness while maintaining the global model's high clean accuracy, modest poisoning rates (usually 10–30%) are employed in previous federated backdoor experiments. This poisoning ratio was selected in accordance with standard settings. A balanced option

that avoids unrealistic full-dataset poisoning and yields high attack success rates under undefended aggregation is a 20% poison rate. To guarantee compatibility with the ResNet-18 input distribution, a 4×4 patch with pixel value 1.0 is added before normalization for CIFAR-10, but a 3×3 trigger is used for MNIST. We validate the attack implementation and assessment pipeline to ensure attack accuracy by confirming that strong poisoning produces near-saturation ASR under FedAvg without protection.

BadNets [49]: A common static backdoor attack known as "BadNets" occurs when malevolent clients reclassify a portion of their local training samples to a predetermined target class by injecting a preset trigger sequence into those samples. By combining poisoned samples with clean data during local training, the global model is able to maintain excellent clean accuracy while learning a strong trigger–target relationship. Applying the trigger to non-target test samples and calculating the proportion categorized as the target label is how the attack success rate (ASR) is calculated in assessment.

CapsuleBD [12]: Instead of depending just on visible input triggers, CapsuleBD is a parameter-space backdoor attack for federated learning that incorporates harmful behavior within client model updates. In order to embed backdoor information into certain parameter subspaces while preserving benign performance, malicious clients improve their local models. This attack's compliance with diverse model parameters and covert update patterns make it very difficult to detect. Triggered test inputs created in accordance with the attack specification are used to assess ASR.

Mirage [51]: A modern and exemplary backdoor attack, Mirage uses adaptive optimization and in-distribution poisoning to accomplish covert and long-lasting backdoor activation. In contrast to static trigger-based assaults, Mirage seeks to reduce detectability while preserving high attack success rates by aligning tainted samples with the natural data distribution. This assault is used to assess how well protections against contemporary backdoor threats are working and represents more sophisticated and real-

istic hostile behavior.

While attack success rate (ASR) is calculated on triggered test samples that adhere to the corresponding attack criteria, clean accuracy (ACC) is assessed on the standard test set across all attack settings. The next sections give detailed experimental findings under various attack methods and attacker ratios.

6.1.3 Defenses Compared

We analyze the suggested approach against representative baseline and cutting-edge defensive mechanisms that adhere to various design philosophies and threat assumptions in order to thoroughly assess its robustness. Every defense is tested with the same attacker ratios and experimental conditions.

FedAvg [1] (no defense): The baseline aggregation technique, Federated Averaging (FedAvg), has no defensive mechanism. Weighted averaging based on local dataset sizes is how the server collects client updates. FedAvg is known to be extremely susceptible to backdoor and poisoning attacks since it doesn't carry out any anomaly detection, client screening, or robustness enhancement. It is included to confirm that the attack implementation is accurate and to set a maximum limit on the attack's efficacy.

Fltrust [29]: FLTrust is a trust-based defense that assesses the dependability of client updates by using a tiny, trusted dataset that is kept up to current by the server. The cosine similarity between each client update and a reliable reference update retrieved from the root dataset is calculated by the server in each round. Prior to aggregation, client contributions are reweighted based on their similarity ratings. FLTrust is not appropriate for rigorous black-box scenarios as it requires access to model updates and presumes the server has clean, reliable data. On the other hand, under white-box or semi-black-box assumptions, it offers a solid foundation.

RFOut [41]: By removing erroneous parameter updates from the model update space, RFOut is a strong aggregation technique that prevents backdoor assaults. RFOut

suppresses suspicious parameter deviations during aggregation using coordinate-wise robust statistics rather than eliminating whole clients. This method assumes direct access to client model parameters but does not call for explicit client categorization or reliable data. The goal of parameter-level robust aggregation protections, such as RFOut, is to restrict the impact of malicious updates while maintaining benign performance.

Proposed Method: Under the rigorous black-box federated learning assumption, the suggested approach, called BLACKBOXGUARD, only sees prediction probability outputs from client models that have been tested on a small private reference dataset. It uses entropy-guided weighted aggregation after round-level client screening with entropy-based behavioral analysis in conjunction with median and MAD statistics. The suggested method does not rely on reliable training data, client data, gradients, logits, or model parameters, in contrast to previous defenses. Strong privacy and deployment compatibility are maintained in actual black-box federated setups, and resistance against backdoor assaults is made possible by this architecture.

6.2 Metrics

Using a common set of criteria, we assess the overhead, robustness, and efficacy of the baseline defenses and the suggested approach. Depending on the goal of the analysis, all metrics are calculated at the server side and given either as final numbers or per round.

Clean Accuracy (ACC): The global model's classification performance on a clean test set devoid of trigger patterns is measured by clean accuracy. It shows if the aggregation mechanism maintains typical learning behavior and represents the usefulness of the learnt model.

$$\mathrm{ACC} = \frac{1}{|D_{\text{clean}}|} \sum_{(x,y)\in D_{\text{clean}}} \mathbb{I}\big(f(x) = y\big). \tag{6.1}$$

where D_{clean} indicates the clean test dataset, $f(\cdot)$ stands for the global model, and $\mathbb{I}(\cdot)$ is the indicator function.

Attack Success Rate (ASR): Attack success rate measures the percentage of trigger-modified inputs that are incorrectly categorized into a target label given by the attacker, hence quantifying the efficacy of a backdoor attack.

$$\mathrm{ASR} = \frac{1}{|D_{\text{trigger}}|} \sum_{(x,y)\in D_{\text{trigger}}} \mathbb{I}\big(f(T(x)) = y^*\big). \tag{6.2}$$

where D_{trigger} is the triggered test set, $T(\cdot)$ represents the trigger injection function, and y^* is the target label. A successful backdoor attack is indicated by a high ASR.

Benign Drop Rate: By measuring the percentage of benign clients that are filtered out by the defense mechanism, the benign drop rate quantifies the false-positive behavior of a defense.

$$\mathrm{BenignDrop} = \frac{\#(\text{benign clients filtered})}{\#(\text{benign clients selected})} \tag{6.3}$$

To reflect temporary benign filtering activity, this measure is assessed at the round level for the suggested approach. ”Benign drop,” as used in the final defensive comparison, refers to the permanent exclusion of benign customers throughout training rounds. The suggested approach has a 0% final benign drop rate, as it only carries out round-level, reversible screening and does not keep client blacklists. As a consequence, no benign client is permanently eliminated.

Server Queries: Server queries measure a defense’s communication and inference costs. In particular, the number of forward passes the server does while assessing client models on a reference dataset is correlated with the number of server requests. The number of server queries for a particular round is about $|\mathcal{S}_t| \times |V|$, where V is the server-side reference dataset and $\mathcal{S}_t$ represents the set of participating clients in round t. There are no server inquiries for defenses that don’t need server-side inference.

Computation Overhead: By breaking down the server-side calculation cost into processing time unique to defense and inference, we further report it. (i) The inference

time t_{infer} needed to calculate prediction probabilities on the reference dataset and (ii) the computation time t_{entropy} for entropy calculation and median/MAD-based screening comprise the overall overhead. The equivalent overhead is nil by design for defenses that do not use entropy-based analysis or server-side inference.

When combined, these measures offer a thorough assessment of system overhead, attack resilience, false-positive behavior, and model utility across various attack scenarios and attacker ratios.

6.3 Evaluation

We first analyze the behavior of the proposed method under controlled settings on MNIST, followed by a comparative evaluation against baseline defenses on CIFAR-10 under different backdoor attacks and attacker ratios.

The MNIST results for the suggested technique are shown in Table 6.1 and Figure 6.1. The entropy-based screening and aggregation approach does not impair benign learning performance, as evidenced by the clean accuracy's quick convergence and stability over 99% during training. On the other hand, the attack success rate (ASR) is continuously reduced to less than 9%, indicating that the backdoor attack is effectively mitigated.

As a result of temporary benign filtering brought on by entropy changes during training, the benign drop rate varies from round to round. Crucially, no benign customer is ever eliminated, and these variations do not build up over time. The ablation findings in Figure 6.1 further demonstrate that the backdoor cannot be completely suppressed by either entropy-weighted aggregation or entropy-based filtering alone. While weighting alone eventually crumbles under constant attack pressure, filtering alone momentarily lowers ASR but is unable to maintain robustness. The only design that continuously maintains low ASR while maintaining excellent clean accuracy is the complete

BLACKBOXGUARD design.

The statistical properties of the entropy-based screening mechanism are in line with the observed variance in the benign drop rate across datasets. Because of MNIST's high prediction certainty and decreased dataset complexity, the entropy distribution across benign clients is tightly grouped. Consequently, slight benign filtering (0.33) may occur when minor oscillations around the median entropy periodically surpass the MAD-based threshold, without compromising the overall usefulness of the model. On the other hand, CIFAR-10 shows a greater inherent prediction variance, which results in a more distinct distinction between malicious and benign entropy profiles. This maintains excellent attack suppression, produces zero benign drop, and enhances threshold discrimination.

Comparative findings on CIFAR-10 under BadNets, CapsuleBD, and Mirage assaults with 30% and 50% malicious involvement are shown in Tables 6.2–6.4. FedAvg maintains great clean accuracy, but shows very high ASR, frequently above 98%, across all attack settings. Given that FedAvg aggregates client updates mindlessly and does not carry out any robustness tests or anomaly mitigation, this behavior is to be anticipated. As a result, the global model completely incorporates malicious updates that include backdoor behavior, enabling the backdoor to take control of the learnt decision function. FedAvg consistently yields the highest ASR across all attacks, confirming the attack pipeline is correctly implemented

FLTrust and RFOut lower ASR in comparison to FedAvg, but they are less resilient to more covert assaults and higher attacker ratios. Specifically, under CapsuleBD and Mirage, if malicious activity is incorporated into minor update patterns or matched with the data distribution, their efficacy deteriorates. According to these findings, safeguards against sophisticated backdoor techniques that just depend on parameter-space robustness or trusted data are inadequate.

On the other hand, the suggested approach maintains competitive clean accuracy

while consistently achieving lower ASR across all attack types and attacker ratios. For instance, even at 50% malicious involvement, ASR is decreased to about 10% under BadNets and CapsuleBD, while the suggested approach significantly outperforms baseline protections under Mirage. Notably, no CIFAR-10 configuration exhibits persistent benign client exclusion, demonstrating that resilience is attained without compromising long-term model utility or client involvement.

Increased server requests and entropy-based computation time are indicative of the additional server-side cost associated with the suggested method's robustness advantages. This overhead, however, has a limit and increases linearly with the server-side reference dataset's size. Cost metrics such as server queries, t_{infer}, and t_{entropy} are reported only for defenses that require server-side inference on client models; for other defenses, these metrics are marked as non-applicable ("–"). Under stringent black-box limitations, the suggested approach provides a positive robustness-cost trade-off when compared to baseline defenses that have little overhead but are unable to stop sophisticated backdoor assaults.

6.4 Ablation Study

In BLACKBOXGUARD, we specifically isolate each component's contribution and assess its impact using the same MNIST federated environment. Figure 6.1 (bottom) reports the ablation, when all variations are trained for the same number of rounds using the same attack setting and data partitioning. **(i) No defense**, **(ii) Entropy-based screening only** (median/MAD filtering without entropy-guided weighting), **(iii) Entropy-weighted aggregation only** (entropy weights without MAD-based screening), and **(iv) Full BLACKBOXGUARD** (screening + weighting) are the variants that we analyze.

The model can learn the benign task, and observed robustness variations are mostly

due to defensive mechanisms rather than unstable training, as demonstrated by the high and consistent clean accuracy across all versions (Figure 6.1, top). The entire BLACKBOXGUARD, in instance, maintains clean accuracy over 99% during training, proving that combining weighting and screening does not compromise benign usefulness.

By eliminating abnormal clients in each round, the screening-only variation lowers ASR in early rounds as compared to the undefended baseline, suggesting that robust entropy screening can inhibit some harmful effects. However, Figure 6.1 (bottom) demonstrates that screening alone is insufficient for long-term robustness: backdoor influence can reappear when malicious behavior partially overlaps the benign entropy distribution or when round-to-round entropy variation weakens the separability under the MAD threshold, and the ASR does not consistently remain low over long horizons. This suggests that screening is insufficient to consistently stop a persistent attacker from progressively altering the global model.

Because lower-entropy clients are given bigger aggregate weights, which lowers the proportional contribution of suspicious clients, the weighting-only alternative likewise initially offers no advantage. Weighting alone, however, eventually fails under persistent attack pressure, as Figure 6.1 (bottom) demonstrates. This is because malicious clients are not specifically filtered out, so their updates continue to be included in aggregation each round, and cumulative backdoor influence can eventually take over the global updates. This behavior demonstrates that reweighting is ineffective to combat persistent backdoor training in the absence of strong screening.

Throughout the entire training procedure, the whole approach consistently keeps the ASR below 9% while maintaining excellent clean accuracy (Table 6.1 and Figure 6.1). The reason for this improvement is that the two elements target complementary failure modes: entropy-guided weighting further lessens the residual impact of borderline clients that stay under the screening threshold, while median/MAD screening eliminates outliers at the behavioral level. Therefore, the ablation confirms that resilience is

due to the combined impact of (i) robust entropy-based screening and (ii) entropy-aware aggregation within the retained client set, rather than a single design decision.

Overall, the ablation findings directly support the BLACKBOXGUARD design reasoning and meet the criteria to show that each component makes a substantial contribution to robustness under the rigorous black-box constraint.

Table 6.1: Evaluation Results on **MNIST** Dataset

Round	**Clean Acc. (%)**	**ASR (%)**	**Benign Drop Rate**	**Server Queries**	t_{infer} **(s) /** $t_{\text{entropy+MAD}}$ **(s)**
10	98.02	8.90	0.33	20000	0.85 / 0.0014
20	98.52	8.79	0.11	20000	0.83 / 0.0017
40	98.76	8.60	0.20	20000	0.81 / 0.0049
60	98.87	8.60	0.00	20000	0.87 / 0.0020
80	99.08	8.60	0.33	20000	0.86 / 0.0018
100	99.00	8.70	0.20	20000	0.83 / 0.0014
120	99.05	8.70	0.11	20000	0.80 / 0.0047
140	99.02	8.79	0.11	20000	0.80 / 0.0049
160	99.06	8.90	0.142	20000	0.82 / 0.0017
180	99.09	8.90	0.33	20000	0.86 / 0.0021
200	**99.15**	**8.90**	0.22	20000	0.82 / 0.0015

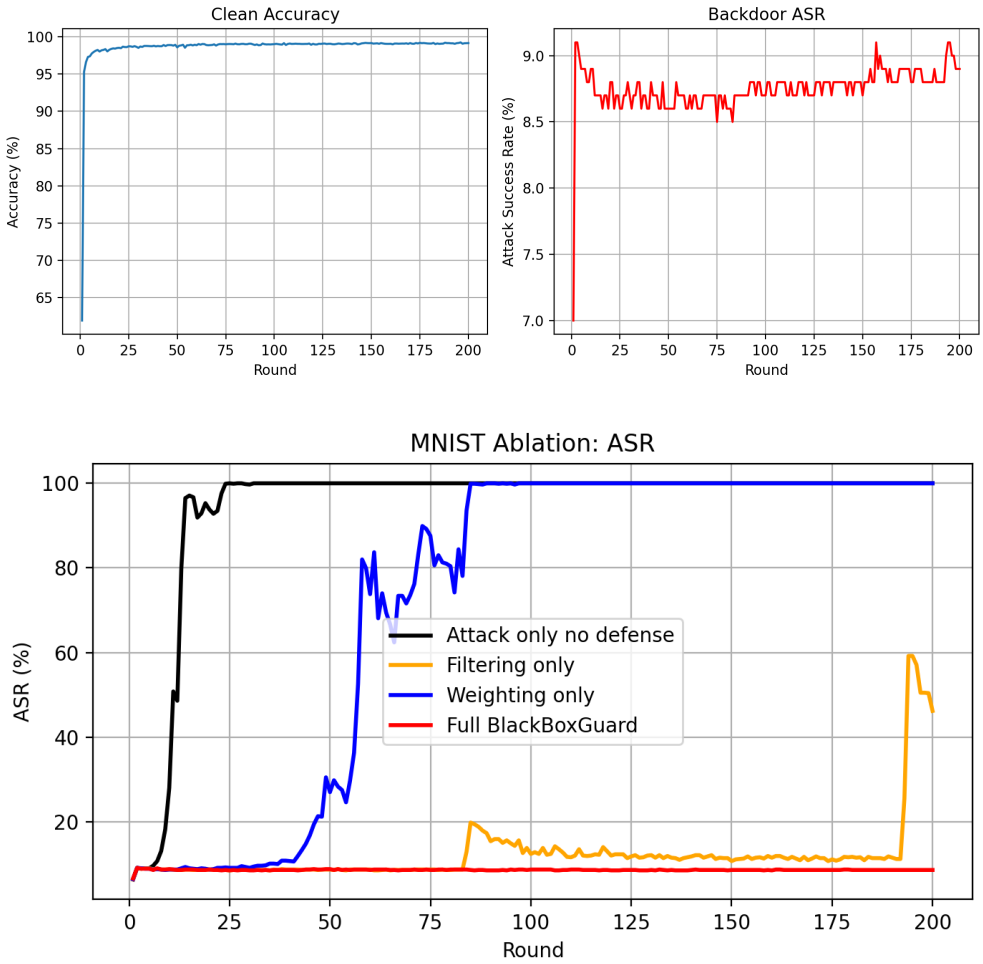

Figure 6.1: MNIST results. **Top:** Clean accuracy and backdoor attack success rate over federated rounds. **Bottom:** ASR comparison under different defense configurations (no defense, filtering only, weighting only, and full BlackBoxGuard).

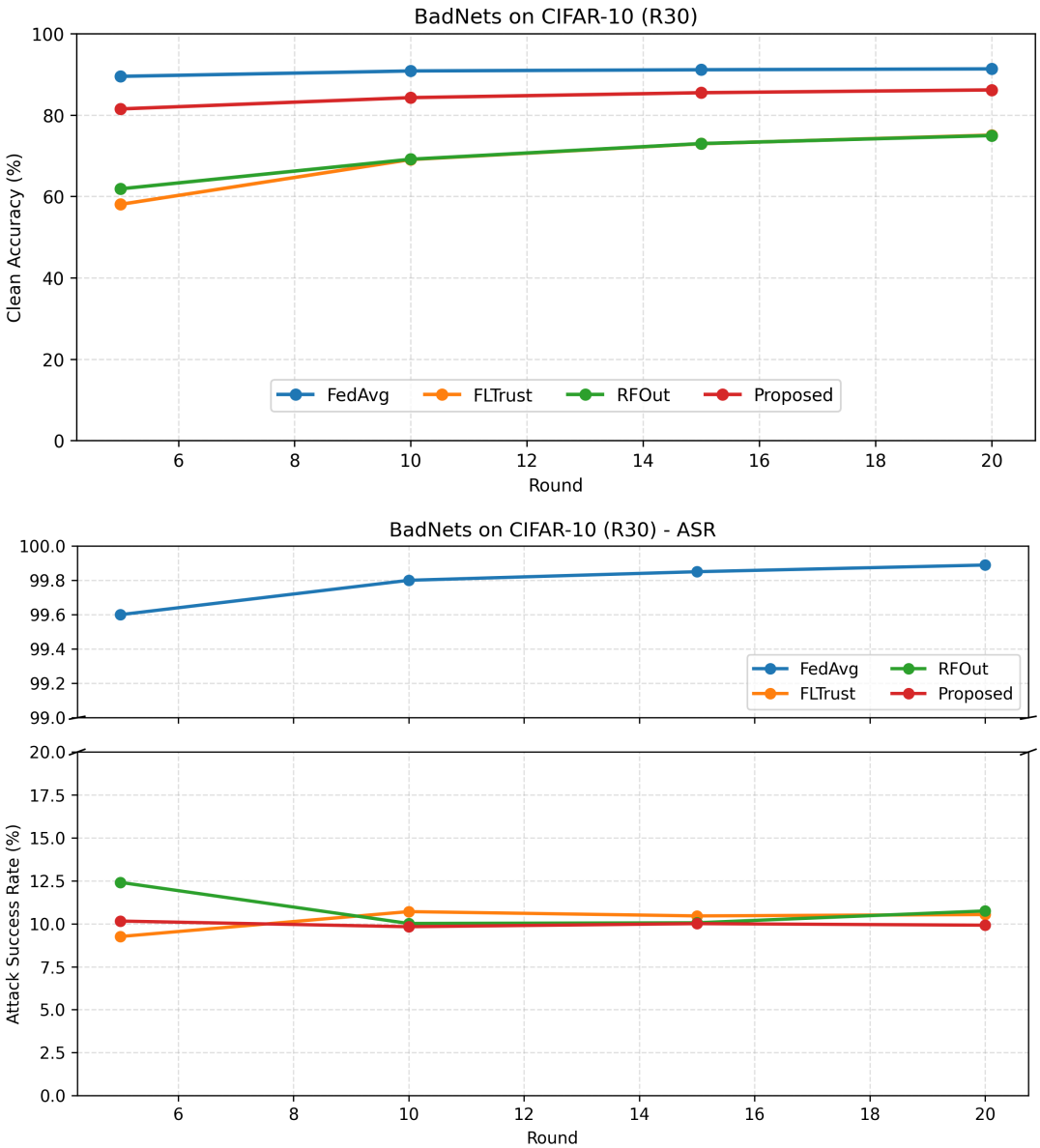

Figure 6.2: Badnets on CIFAR-10 (30% malicious clients): clean accuracy (top) and attack success rate (bottom).

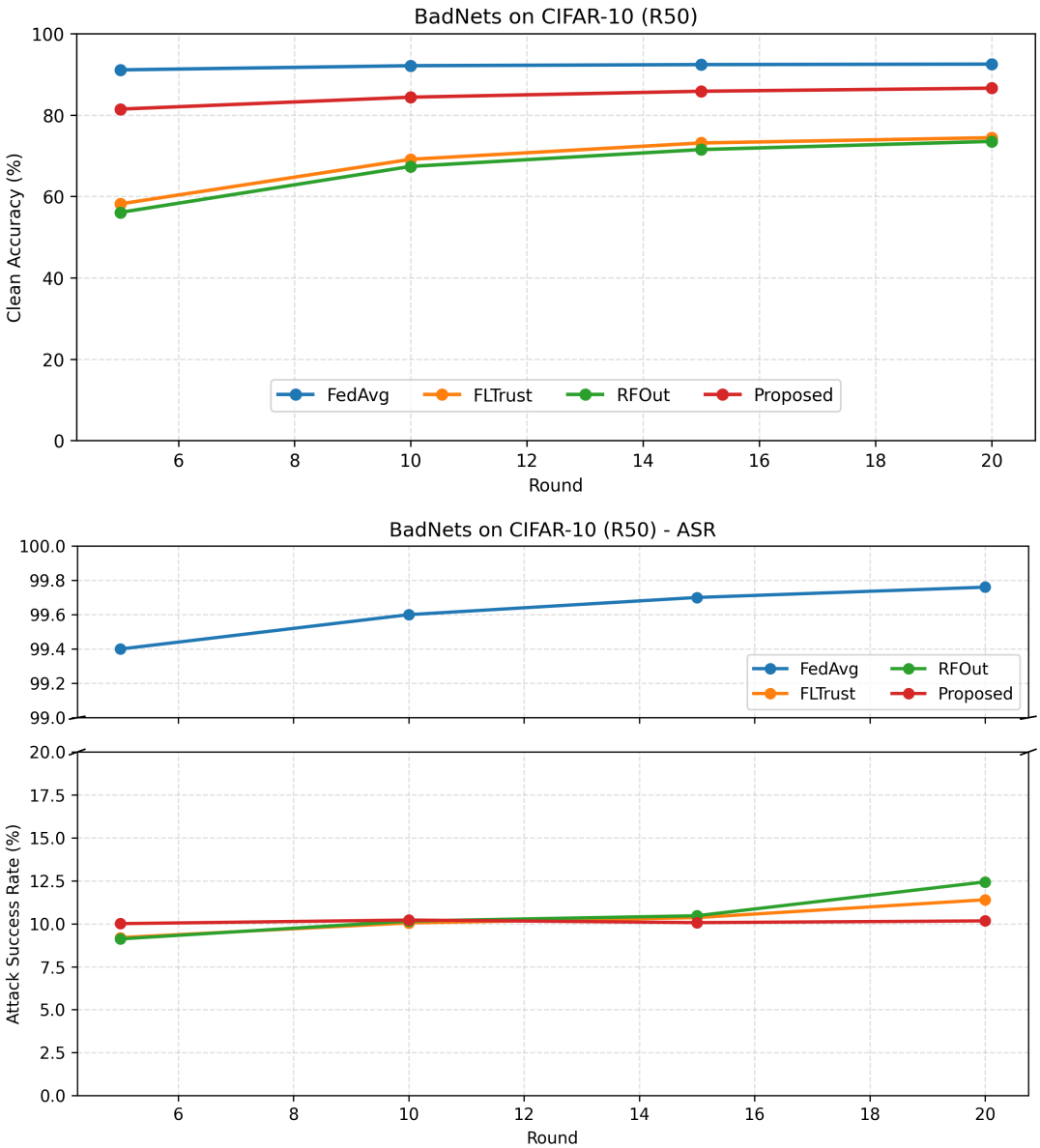

Figure 6.3: Badnets on CIFAR-10 (50% malicious clients): clean accuracy (top) and attack success rate (bottom).

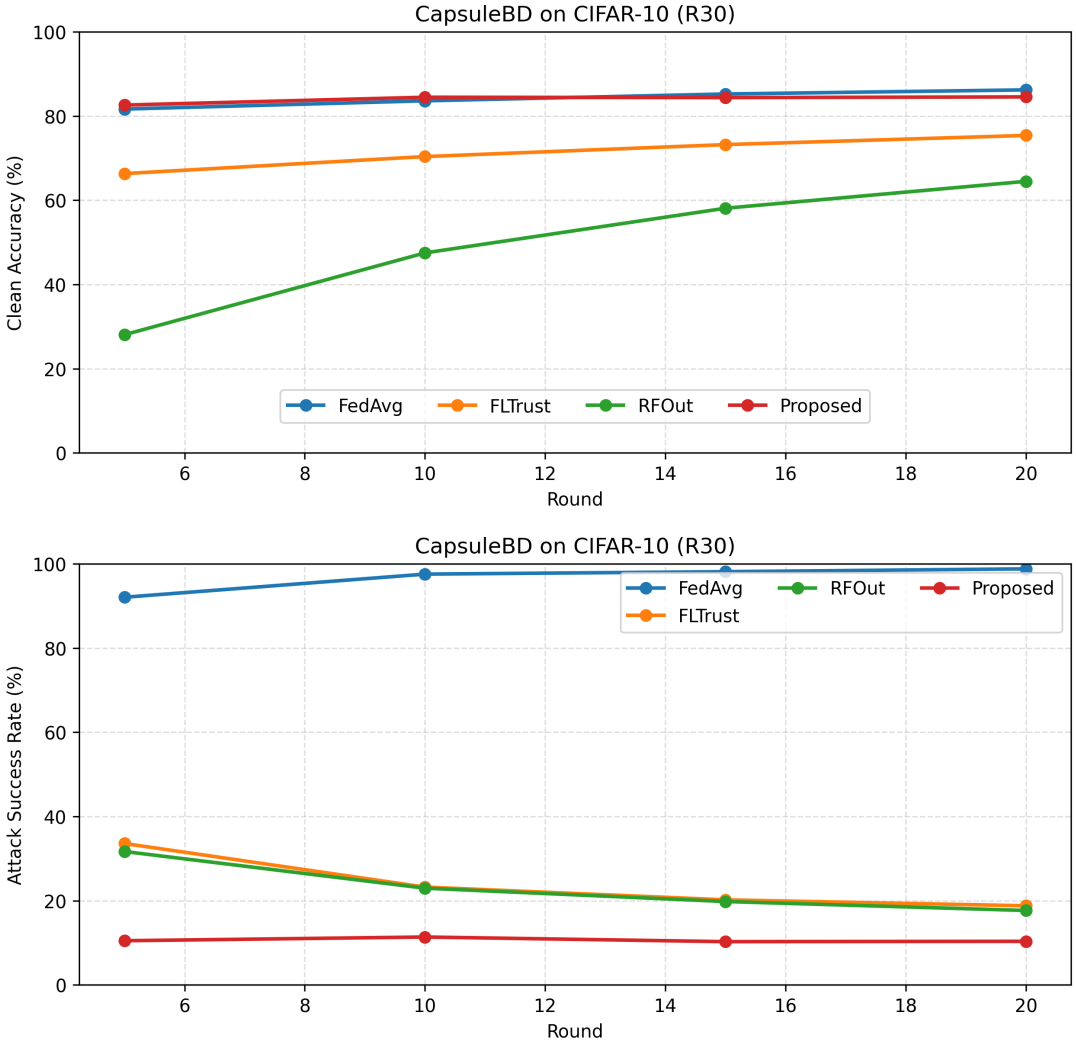

Figure 6.4: CapsuleBD on CIFAR-10 (30% malicious clients): clean accuracy (top) and attack success rate (bottom).

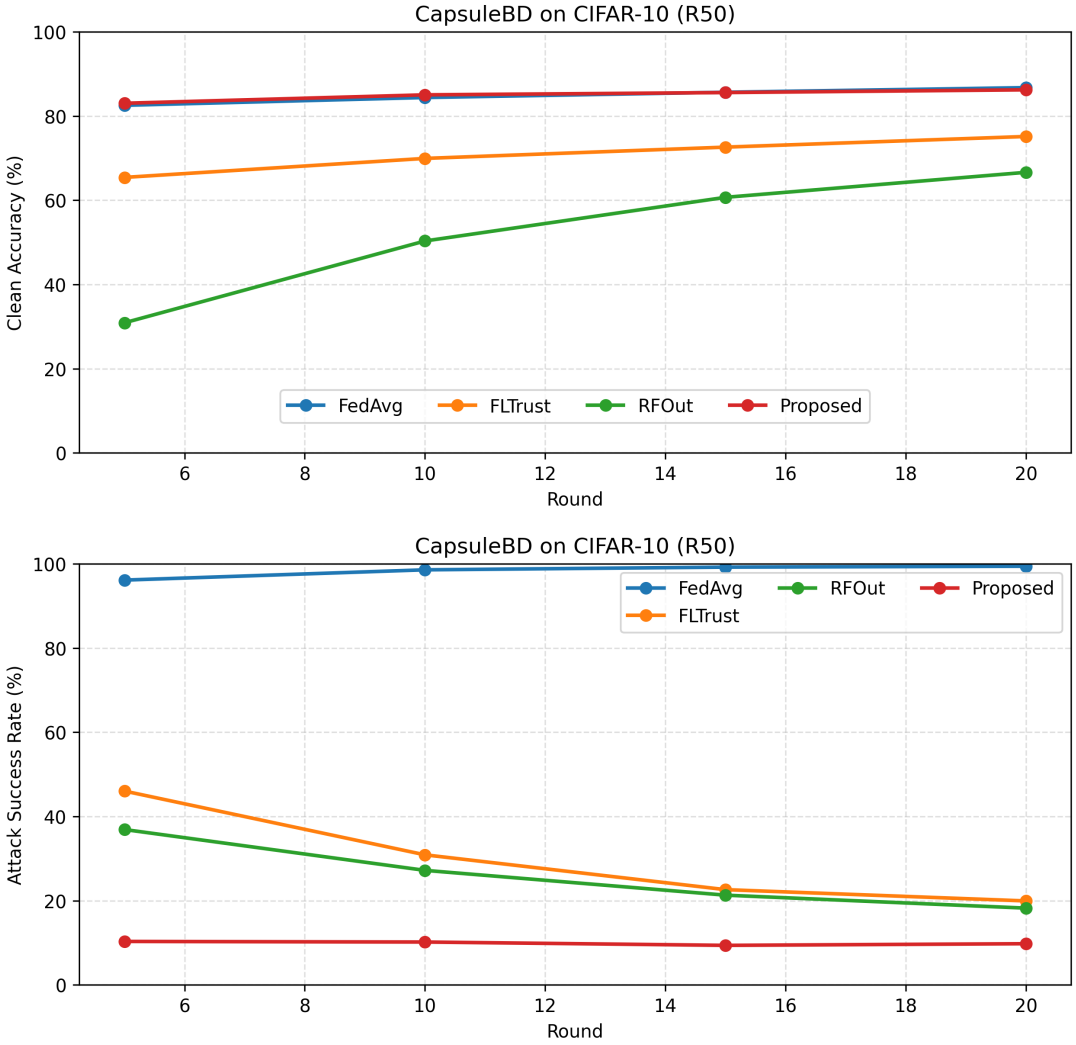

Figure 6.5: CapsuleBD on CIFAR-10 (50% malicious clients): clean accuracy (top) and attack success rate (bottom).

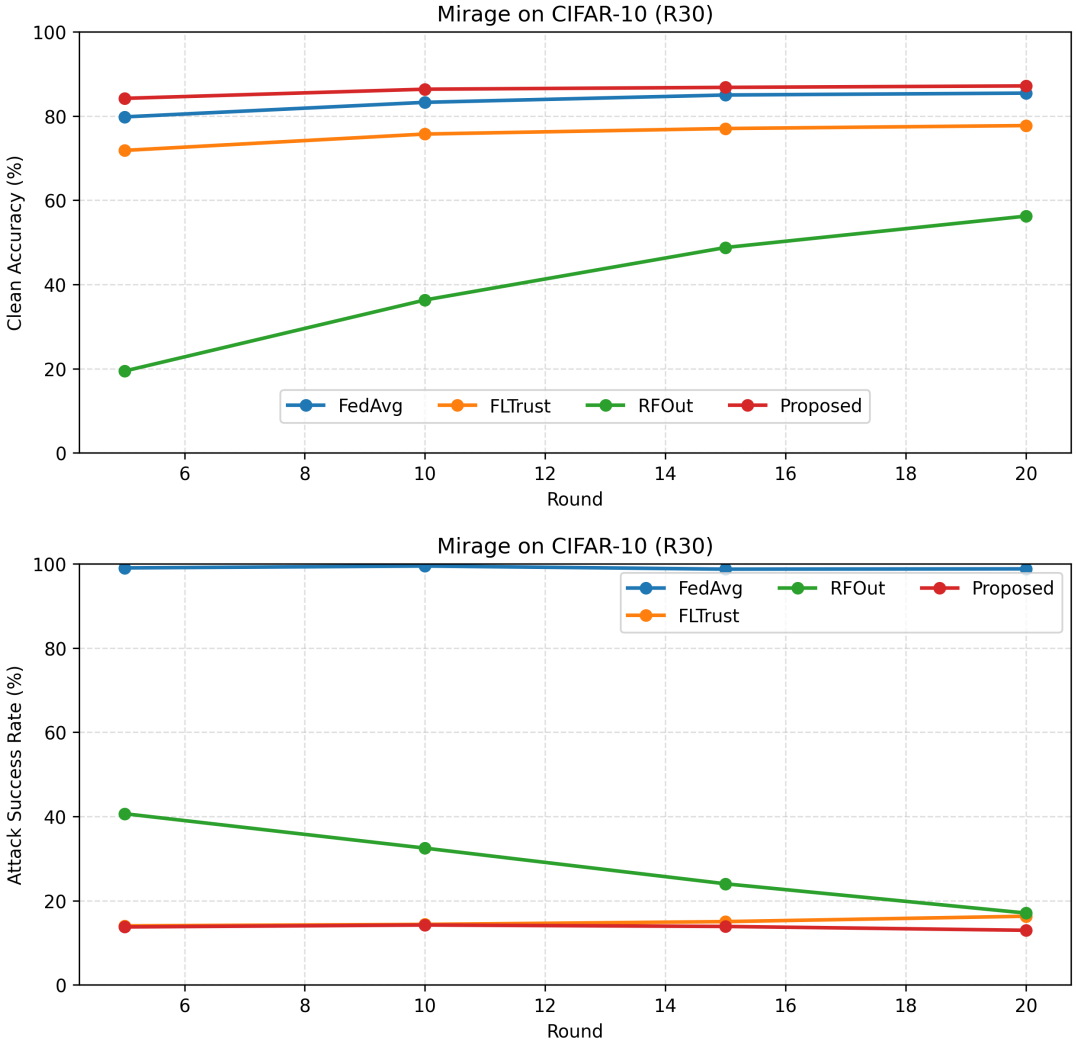

Figure 6.6: Mirage on CIFAR-10 (30% malicious clients): clean accuracy (top) and attack success rate (bottom).

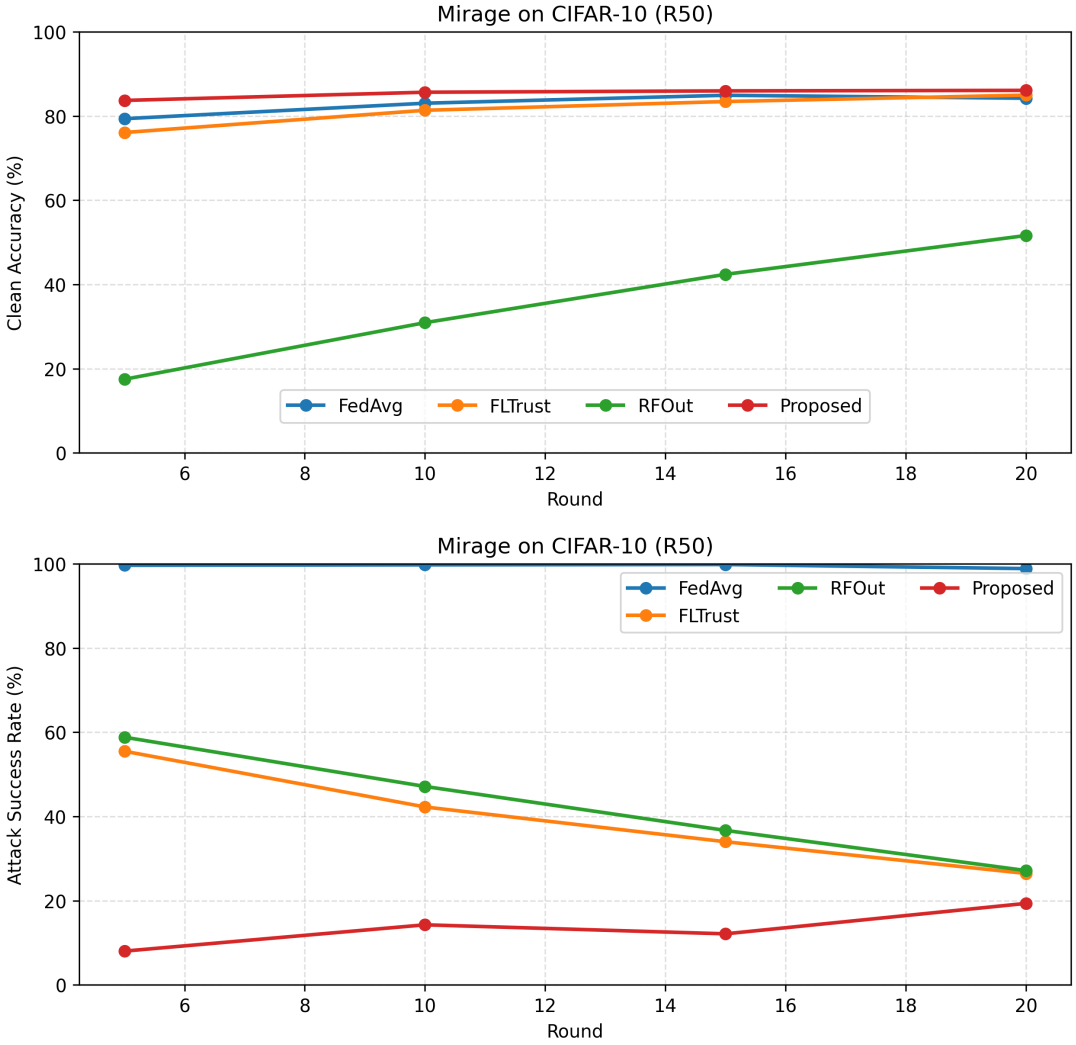

Figure 6.7: Mirage on CIFAR-10 (30% malicious clients): clean accuracy (top) and attack success rate (bottom).

Table 6.2: Comparison of aggregation-based defenses under the BadNets attack on CIFAR-10 (30% and 50% malicious clients).

Attack	**Metric**	**FedAvg**		**FLTrust**		**RFOut**		**Proposed**	
		R30	R50	R30	R50	R30	R50	R30	R50
BadNets	ACC	91.34	92.51	75.06	74.43	74.96	73.52	86.17	86.6
	ASR	99.88	99.76	10.54	11.4	10.75	12.44	9.92	10.17
	Benign Drop	-	-	-	-	-	-	0.0	0.0
	Server Queries	-	-	-	-	-	-	6000	8000
	t_{infer}	-	-	-	-	-	-	2.80	3.94
	t_{entropy}	-	-	-	-	-	-	0.00016	0.00015

Table 6.3: Comparison of aggregation-based defenses under the CapsuleBD attack on CIFAR-10 (30% and 50% malicious clients).

Attack	**Metric**	**FedAvg**		**FLTrust**		**RFOut**		**Proposed**	
		R30	R50	R30	R50	R30	R50	R30	R50
CapsuleBD	ACC	86.21	86.74	75.39	75.14	64.49	66.63	84.54	86.22
	ASR	98.8	99.42	18.78	19.93	17.66	18.22	10.34	9.77
	Benign Drop	-	-	-	-	-	-	0	0
	Server Queries	-	-	-	-	-	-	6000	8000
	t_{infer}	-	-	-	-	-	-	3.62	4.39
	t_{entropy}	-	-	-	-	-	-	0.00017	0.00018

Table 6.4: Comparison of aggregation-based defenses under the Mirage attack on CIFAR-10 (30% and 50% malicious clients).

Attack	**Metric**	**FedAvg**		**FLTrust**		**RFOut**		**Proposed**	
		R30	R50	R30	R50	R30	R50	R30	R50
Mirage	ACC	85.44	84.22	77.73	84.96	56.23	51.62	87.15	86.09
	ASR	98.8	98.86	16.30	26.45	17.05	27.14	12.96	19.35
	Benign Drop	-	-	-	-	-	-	0	0
	Server Queries	-	-	-	-	-	-	6000	8000
	t_{infer}	-	-	-	-	-	-	10.99	4.53
	t_{entropy}	-	-	-	-	-	-	0.00019	0.00017

Chapter 7

Security Analysis

The security aspects of the proposed entropy-based prediction filtering mechanism (BlackBoxGuard) within the federated learning threat model covered in this thesis are examined in this part. The analysis highlights the method's resistance to backdoor assaults in a completely black-box environment, where the server relies solely on prediction outputs and has no control over client gradients, network changes, or internal parameters. This section seeks to outline the security assurances offered by the proposed approach and to support its efficacy within the constraints of realistic deployment.

7.1 Threat Model and Backdoor Robustness

This study centers on a typical federated learning framework comprising a central server and numerous distributed clients, out of which some may act maliciously. These hostile clients execute backdoor attacks by introducing poisoned samples that contain a trigger pattern into their local training datasets, attempting to alter the global model to misclassify triggered inputs as a specified label selected by the attacker, while keeping high accuracy on pristine data [9, 12].

Malicious activity is not limited to a single backdoor construction in this secu-

rity investigation. Rather, the threat model includes many attack realizations that are assessed in this work, such as heterogeneous client-side backdoor injection, adaptive optimization-driven poisoning, and static trigger insertion. These assault configurations match the Mirage, CapsuleBD, and BadNets attack models examined in the experimental evaluation, respectively. All of them are regarded as realistic adversaries under the same federated learning assumptions, notwithstanding the differences in their implementation methodologies.

Under a rigorous black-box threat model, it is believed that client-side gradients are not accessible to the server, model parameters, or update vectors. Instead, it can solely observe the prediction outputs produced by client models when assessed on a limited validation dataset managed by the server, a premise that aligns with realistic privacy-protecting implementations [7]. Given this limitation, the proposed method assesses client behavior by examining prediction uncertainty through entropy, which measures the confidence distribution of model outputs. Backdoored models typically show unusually low entropy for triggered inputs due to their overconfident misclassification tendencies.

By aggregating entropy statistics from all participating clients and employing robust statistical filtering based on the median and median absolute deviation (MAD), the server can detect abnormal prediction behaviors that signal backdoor activity [53, 54]. Without requiring access to internal model data, this screening method reduces the impact of fraudulent clients during aggregation. The suggested approach successfully lowers the success of backdoor assaults while maintaining high clean accuracy during federated learning rounds, as demonstrated by experimental results on MNIST and CIFAR-10.

7.2 Security Properties and Limitations

In a strictly black-box federated learning scenario, where the server can only view predicted results produced by the client systems and has no control over client-side variables, parameters for the model, modification vectors, or its own training states, the suggested entropy-based prediction filtering mechanism offers resilience against backdoor attacks. The security features provided by the suggested method are outlined in this part, together with an explanation of the underlying presumptions and restrictions that underlie these assurances.

The suggested method's security is predicated on the benign-majority assumption, which states that less than half of the clients involved in each communication round act maliciously. In this scenario, most customers show patterns of prediction uncertainty that are consistent with benign optimization goals. Reliable identification of entropy variations induced by backdoor training is made possible by robust statistical metrics like the median and the Median Absolute Deviation (MAD), which remain steady in the face of a minority of anomalous values. This premise is essential to a wide class of Byzantine-robust aggregation and anomaly detection methods in federated learning [25, 53], and it is not specific to the suggested approach.

Entropy-based screening efficiently reduces the impact of malicious clients when the benign-majority condition is fulfilled without requiring access to gradients, model parameters, or trusted reference updates. Because internal client information cannot be disclosed in privacy-sensitive federated learning deployments, the suggested approach achieves robustness under much weaker observability assumptions than many existing defenses [7].

The statistical basis of the defense is no longer valid if over 50% of the participating clients are malevolent. In certain situations, the distribution may be dominated by malevolent entropy patterns, which would change the median and make deviation-based detection useless. This constraint is essential to median-based and majority-

driven robust statistics and applies equally to various Byzantine-robust aggregation approaches [25]. Practically speaking, majority adversarial control scenarios are typically regarded as outside of realistic federated learning threat models because they suggest systemic flaws in client enrollment, authentication, or trust management rather than deficiencies in aggregation or defense mechanisms. As a result, the suggested approach does not seek to offer assurances under majority malevolent involvement, much like current strong defenses.

The suggested method shows robustness against various backdoor attack techniques inside the benign-majority regime. Successful backdoor injection necessitates very confidence misclassification toward attacker-defined target labels, even when adversaries may differ in poisoning rates, trigger designs, or optimization behaviors. Distorted prediction uncertainty profiles are a behavioral consequence that may be seen at the server even in the absence of internal model information [8, 49].

Due of the rigid black-box setup, the suggested approach has additional drawbacks. It is predicated on the server having a small, clean validation dataset available for behavioral evaluation. Furthermore, the efficacy of detection may be diminished by highly adaptable adversaries that meticulously imitate benign entropy patterns on clean data. In black-box federated learning settings, these restrictions represent a fundamental trade-off between privacy preservation and defensive capabilities, especially under sophisticated adaptive threat models [24].

From a security standpoint, the suggested defense is independent of presumptions unique to a certain backdoor technique. The studied attacks—ranging from fixed-pattern poisoning to optimization-driven and heterogeneous backdoor behaviors—exert impact on the global model through different techniques. However, for a backdoor attack to be successful, systematic deviations in prediction behavior must be introduced and maintained during training sessions. The efficacy of the suggested approach across the many attack scenarios investigated can be explained by the fact that it makes use of this shared

consequence rather than the assault architecture itself.

In the end, the security analysis shows that, under realistic federated learning assumptions, the suggested entropy-based prediction filtering mechanism provides strong and useful robustness against backdoor attacks while openly recognizing the inherent limitations imposed by stringent black-box observability and benign-majority requirements.

7.2.1 Mimicking Entropy and Adaptive Adversaries

A malicious client may try to mimic benign behavior by making accurate predictions on server-side validation samples, which might be a problem for entropy-based security. Instantaneous detection is difficult in these situations because entropy levels may momentarily approximate those of honest customers. Given that attackers can carefully strike a compromise between attack efficacy and stealth in federated learning, this problem is very pertinent. Nevertheless, the attacker must continuously impose confidence misclassification toward a target label over training rounds in order to maintain a successful backdoor assault. Without weakening the backdoor itself, it is impossible to remove the systematic distortions in prediction uncertainty introduced by this persistent goal. Because of this, it is challenging to maintain long-term entropy alignment with benign clients under successful backdoor injection, even when short-term entropy imitation is feasible.

Experimental results across various backdoor tactics, such as optimization-driven assaults (mirage) [51], fixed trigger-based attacks (BadNets) [9], and heterogeneous backdoor mechanisms (CapsuleBD) [12], empirically represent this trade-off. Entropy-based screening may function successfully under a variety of threat models since all examined assaults show persistent confidence manipulation when the backdoor is still functional, despite their different attack aims and implementations. We stress that if an attacker maintains a substantial backdoor effect and fully matches benign predic-

tion uncertainty, no prediction-only strategy can ensure discovery. All defenses that function without access to gradients, parameters, or reliable reference models have this constraint, which is essential to tight black-box observability. As a result, our strategy targets real-world adversaries rather than theoretically ideal attackers and assumes a benign majority.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

In a rigorous black-box environment, where the server can only view model prediction outputs and cannot access gradients, model parameters, or client-side training data, this thesis tackled the problem of protecting federated learning systems from backdoor assaults. This study focuses on a realistic deployment situation molded by privacy limitations, secure computing, and restricted observability, in contrast to many prior defenses that rely on trusted reference updates or internal model visibility. An entropy-based prediction filtering system was proposed to define client behavior through prediction uncertainty in order to meet this scenario. The suggested method allows the server to detect and reduce fraudulent contributions using just observable outputs by utilizing entropy as a behavioral cue. This approach highlights the usefulness of prediction-level analysis in federated learning contexts by showing that efficient backdoor mitigation is possible without going against black-box assumptions. One important finding of this study is that the behavioral consequence of backdoor assaults is impaired prediction confidence, regardless of how they are implemented. The suggested approach can generalize across various backdoor tactics without depending on attack-specific signatures

or gradient-based heuristics since entropy captures this impact in an attack-agnostic manner. Because of this, entropy is a signal that is especially well-suited for reliable client screening when observability is limited. Evaluated against static trigger-based, heterogeneous, and optimization-driven backdoor attacks

The suggested approach also recognizes its own limits. A benign-majority assumption, which is essential to reliable statistical methods based on median-driven analysis, is what makes it effective. Furthermore, highly adaptable adversaries that closely resemble benign prediction uncertainty patterns may pose a threat to the method, which also implies the availability of a small, clean validation dataset stored by the server. In rigorous black-box federated learning, these restrictions represent inevitable trade-offs between privacy protection and defensive capabilities. In general, this research shows that, within practical constraints, prediction-based entropy analysis offers a workable and moral basis for backdoor-resilient federated learning. The suggested method makes it more feasible to implement secure federated learning systems in real-world scenarios where privacy is a concern by doing away with the need for gradients, model updates, or internal representations.

8.2 Future Work

In the entropy-based prediction filtering phase, the suggested approach utilizes fixed validation queries to assess prediction entropy and detect anomalous client behavior. This structure facilitates effective backdoor defense under stringent black-box conditions. Future efforts could optimize this phase by integrating adaptive query selection techniques, enabling the server to modify the validation set according to detected patterns of prediction uncertainty. Such enhancements might increase sensitivity to nuances or evolving backdoor attacks. Although the proposed method shows considerable resilience in moderate federated environments, its effectiveness may diminish as the

training setup becomes more intricate. Specifically, with a growing number of participating clients or more coordinated attacker actions, identifying malicious prediction patterns could prove increasingly difficult. This challenge primarily results from a greater overlap between benign and malicious prediction distributions in expansive federated contexts. Tackling this issue is essential for broadening the method's applicability in practical federated learning systems, where diverse client populations and varied data distributions are typical. Upcoming research will concentrate on refining the statistical modeling of prediction entropy, incorporating temporal consistency throughout training rounds, and enhancing scalability while maintaining black-box principles. These advancements aim to bolster the robustness and utility of the proposed defense across varied and large-scale federated learning settings.

Bibliography

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*. PMLR, 2017, pp. 1273–1282.

[2] L. Lyu, H. Yu, X. Ma, C. Chen, L. Sun, J. Zhao, Q. Yang, and P. S. Yu, "Privacy and robustness in federated learning: Attacks and defenses," *IEEE transactions on neural networks and learning systems*, vol. 35, no. 7, pp. 8726–8746, 2022.

[3] C. Xie, K. Huang, P.-Y. Chen, and B. Li, "Dba: Distributed backdoor attacks against federated learning," in *International conference on learning representations*, 2019.

[4] Z. Zhang, A. Panda, L. Song, Y. Yang, M. Mahoney, P. Mittal, R. Kannan, and J. Gonzalez, "Neurotoxin: Durable backdoors in federated learning," in *International conference on machine learning*. PMLR, 2022, pp. 26 429–26 446.

[5] D. Cajaraville-Aboy, A. Fernández-Vilas, R. P. Díaz-Redondo, and M. Fernández-Veiga, "Byzantine-robust aggregation for securing decentralized federated learning," *IEEE Access*, vol. 13, pp. 190 947–190 963, 2025.

[6] Y. Li, J. Xu, and D. Yang, "Flcom: Robust federated learning against strong model poisoning attacks," *Computer Networks*, vol. 269, p. 111442,

2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1389128625004098

[7] F. Zhao, Y. Xie, X. Ren, B. Ding, S. Yang, and Y. Li, "Understanding byzantine robustness in federated learning with a black-box server," *arXiv preprint arXiv:2408.06042*, 2024.

[8] J. Guo, Y. Li, X. Chen, H. Guo, L. Sun, and C. Liu, "Scale-up: An efficient black-box input-level backdoor detection via analyzing scaled prediction consistency," *arXiv preprint arXiv:2302.03251*, 2023.

[9] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How To Backdoor Federated Learning," 2019. [Online]. Available: https://arxiv.org/abs/1807.00459

[10] X. Lyu, Y. Han, W. Wang, J. Liu, Y. Zhu, G. Xu, J. Liu, and X. Zhang, "Lurking in the shadows: Unveiling stealthy backdoor attacks against personalized federated learning," in *33rd USENIX Security Symposium (USENIX Security 24)*, 2024, pp. 4157–4174.

[11] X. Han, X. Zhang, X. Lan, H. Wang, S. Xu, S. Ren, J. Zeng, M. Wu, M. Heinrich, and T. Zhang, "Badsfl: Backdoor attack against scaffold federated learning," *arXiv preprint arXiv:2411.16167*, 2024.

[12] Y. Liao, X. Zhao, B. Zhou, and Y. Huang, "Capsulebd: A backdoor attack method against federated learning under heterogeneous models," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 4071–4086, 2025.

[13] Y. Qiao, D. Liu, R. Wang, and K. Liang, "Low-frequency black-box backdoor attack via evolutionary algorithm," in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 2025, pp. 7582–7592.

[14] T. Wang, Y. Yao, F. Xu, S. An, H. Tong, and T. Wang, "An invisible black-box backdoor attack through frequency domain," in *European Conference on Computer Vision*. Springer, 2022, pp. 396–413.

[15] H. Li, Q. Ye, H. Hu, J. Li, L. Wang, C. Fang, and J. Shi, "3dfed: Adaptive and extensible framework for covert backdoor attack in federated learning," in *2023 IEEE symposium on security and privacy (SP)*. IEEE, 2023, pp. 1893–1907.

[16] C. Shi, S. Ji, X. Pan, X. Zhang, M. Zhang, M. Yang, J. Zhou, J. Yin, and T. Wang, "Towards practical backdoor attacks on federated learning systems," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 6, pp. 5431–5447, 2024.

[17] X. Gong, Y. Chen, H. Huang, Y. Liao, S. Wang, and Q. Wang, "Coordinated backdoor attacks against federated learning with model-dependent triggers," *IEEE network*, vol. 36, no. 1, pp. 84–90, 2022.

[18] H. Zhang, J. Jia, J. Chen, L. Lin, and D. Wu, "A3fl: Adversarially adaptive backdoor attacks to federated learning," *Advances in neural information processing systems*, vol. 36, pp. 61 213–61 233, 2023.

[19] X. Lyu, Y. Han, W. Wang, J. Liu, B. Wang, J. Liu, and X. Zhang, "Poisoning with cerberus: Stealthy and colluded backdoor attack against federated learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 7, 2023, pp. 9020–9028.

[20] T. D. Nguyen, T. A. Nguyen, A. Tran, K. D. Doan, and K.-S. Wong, "Iba: Towards irreversible backdoor attacks in federated learning," *Advances in Neural Information Processing Systems*, vol. 36, pp. 66 364–66 376, 2023.

[21] D. Liu, Y. Qiao, R. Wang, K. Liang, and G. Smaragdakis, "Ladder: Multi-objective backdoor attack via evolutionary algorithm," *arXiv preprint arXiv:2411.19075*, 2024.

[22] K. Y. Yoo and N. Kwak, "Backdoor attacks in federated learning by rare embeddings and gradient ensembling," in *Proceedings of the 2022 conference on empirical methods in natural language processing*, 2022, pp. 72–88.

[23] M. Li, W. Wan, Y. Ning, S. Hu, L. Xue, L. Y. Zhang, and Y. Wang, "Darkfed: A data-free backdoor attack in federated learning," *arXiv preprint arXiv:2405.03299*, 2024.

[24] R. McGaughey, J. M. del Rincon, and I. Alouani, "Stealth by conformity: Evading robust aggregation through adaptive poisoning," *arXiv preprint arXiv:2509.08746*, 2025.

[25] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/f4b9ec30ad9f68f89b29639786cb62ef-Paper.pdf

[26] E. M. E. Mhamdi, R. Guerraoui, and S. Rouault, "The hidden vulnerability of distributed learning in byzantium," in *International Conference on Machine Learning*, 2018. [Online]. Available: https://api.semanticscholar.org/CorpusID:3473997

[27] X. Fang and M. Ye, "Robust federated learning with noisy and heterogeneous clients," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 10 062–10 071.

[28] K. Pillutla, S. M. Kakade, and Z. Harchaoui, “Robust aggregation for federated learning,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 1142–1154, 2022.

[29] X. Cao, M. Fang, J. Liu, and N. Z. Gong, “Fltrust: Byzantine-robust federated learning via trust bootstrapping,” 2022. [Online]. Available: https://arxiv.org/abs/2012.13995

[30] S. Andreina, G. A. Marson, H. Möllering, and G. Karame, “Baffle: Backdoor detection via feedback-based federated learning,” in *2021 IEEE 41st International Conference on Distributed Computing Systems (ICDCS)*, 2021, pp. 852–863.

[31] C. Zhu, J. Zhang, X. Sun, B. Chen, and W. Meng, “Adfl: Defending backdoor attacks in federated learning via adversarial distillation,” *Computers and Security*, vol. 132, 2023.

[32] P. Liu, X. Xu, and W. Wang, “Threats, attacks and defenses to federated learning: issues, taxonomy and perspectives,” *Cybersecurity*, vol. 5, no. 1, p. 4, 2022.

[33] Y. Dong, X. Yang, Z. Deng, T. Pang, Z. Xiao, H. Su, and J. Zhu, “Black-box detection of backdoor attacks with limited information and data,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 16 482–16 491.

[34] Y. Miao, R. Xie, X. Li, Z. Liu, K.-K. R. Choo, and R. H. Deng, “Efficient and secure federated learning against backdoor attacks,” *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 5, pp. 4619–4636, 2024.

[35] J. Paranjape, S. Sikder, S. Vedula, and V. Patel, “Federated black-box adaptation for semantic segmentation,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 112 814–112 835, 2024.

[36] E. Kabir, Z. Song, M. R. U. Rashid, and S. Mehnaz, "Flshield: a validation-based federated learning framework to defend against poisoning attacks," in *2024 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2024, pp. 2572–2590.

[37] Z. Zhang, J. Li, S. Yu, and C. Makaya, "Safelearning: Secure aggregation in federated learning with backdoor detectability," *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 3289–3304, 2023.

[38] H. Zhang, X. Li, Y. Miao, S. Yuan, M. Zhu, X. Liu, and R. H. Deng, "Fl-cdf: Collaborative defense framework for backdoor mitigation in federated learning," *IEEE Transactions on Dependable and Secure Computing*, 2025.

[39] Y. Chen, H. Gao, H. Wang, and T. Gao, "Flsda: A synergistic defense against backdoor attacks in federated learning," *Information Sciences*, p. 122974, 2025.

[40] Z. Qin, F. Chen, C. Zhi, X. Yan, and S. Deng, "Resisting backdoor attacks in federated learning via bidirectional elections and individual perspective," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 13, 2024, pp. 14 677–14 685.

[41] N. Rodríguez-Barroso, E. Martínez-Cámara, M. V. Luzón, and F. Herrera, "Backdoor attacks-resilient aggregation based on robust filtering of outliers in federated learning for image classification," *Knowledge-Based Systems*, vol. 245, p. 108588, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950705122002635

[42] J. Lavond, M. Cheng, and Y. Li, "Trusted aggregation (tag): Backdoor defense in federated learning," *Transactions on Machine Learning Research*, 2024.

[43] E. M. Campos, A. Gonzalez-Vidal, J. L. Hernández-Ramos, and A. Skarmeta, "Fedrdf: A robust and dynamic aggregation function against poisoning attacks in

federated learning," *IEEE Transactions on Emerging Topics in Computing*, vol. 13, no. 1, pp. 48–67, 2025.

[44] G. Yan, H. Wang, X. Yuan, and J. Li, "Fedrola: Robust federated learning against model poisoning via layer-based aggregation," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24)*, 08 2024, pp. 3667–3678.

[45] Y. Siriwardhana, P. Porambage, M. Liyanage, S. Marchal, and M. Ylianttila, "Robust aggregation technique against poisoning attacks in multi-stage federated learning applications," in *2024 IEEE 21st Consumer Communications Networking Conference (CCNC)*, 2024, pp. 956–962.

[46] R. Zhang, Y. Zhang, Y. Zhao, B. Jia, and W. Lian, "Fedcvg: a two-stage robust federated learning optimization algorithm," *Scientific Reports*, vol. 15, no. 1, p. 18357, 2025.

[47] C. E. Shannon, "A mathematical theory of communication," *The Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.

[48] J. Ibrahim and S. Gajin, "Entropy-based network traffic anomaly classification method resilient to deception," *Computer Science and Information Systems*, vol. 19, no. 1, pp. 87–116, 2022.

[49] T. Gu, B. Dolan-Gavitt, and S. Garg, "Badnets: Identifying vulnerabilities in the machine learning model supply chain," *arXiv preprint arXiv:1708.06733*, 2017.

[50] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," *CoRR*, vol. abs/1706.04599, 2017. [Online]. Available: http://arxiv.org/abs/1706.04599

[51] Y. Li, Y. Zhao, C. Zhu, and J. Zhang, "Infighting in the dark: Multi-label backdoor attack in federated learning," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 25 770–25 779.

[52] P. Fang and J. Chen, "On the vulnerability of backdoor defenses for federated learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 10, 2023, pp. 11 800–11 808.

[53] M. A. Romo-Chavero, J. A. Cantoral-Ceballos, J. A. Pérez-Díaz, and C. Martinez-Cagnazzo, "Median absolute deviation for bgp anomaly detection," *Future Internet*, vol. 16, no. 5, p. 146, 2024.

[54] G. Singh and S. Kundu, "Outlier and trend detection using approximate median and median absolute deviation," in *2022 5th International Conference on Computational Intelligence and Networks (CINE)*. IEEE, 2022, pp. 01–06.

[55] J. Liu, C. Peng, W. Tan, and C. Shi, "Federated learning backdoor attack based on frequency domain injection," *Entropy*, vol. 26, no. 2, p. 164, 2024.

[56] T. Liu, Y. Zhang, Z. Feng, Z. Yang, C. Xu, D. Man, and W. Yang, "Beyond traditional threats: A persistent backdoor attack on federated learning," 2024. [Online]. Available: https://arxiv.org/abs/2404.17617

[57] S. Li, E. Ngai, and T. Voigt, "Byzantine-robust aggregation in federated learning empowered industrial iot," *IEEE Transactions on Industrial Informatics*, vol. 19, no. 2, pp. 1165–1175, 2023.

[58] L. Muñoz-González, K. T. Co, and E. C. Lupu, "Byzantine-robust federated machine learning through adaptive model averaging," *arXiv preprint arXiv:1909.05125*, 2019.

[59] X. Yan, L. Wu, Z. Zhang, B. Liu, L. Huo, and J. Wang, "Fedbap: Backdoor defense via benign adversarial perturbation in federated learning," in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 11 947–11 955.

[60] T. Huang, S. Hu, K.-H. Chow, F. Ilhan, S. Tekin, and L. Liu, "Lockdown: backdoor defense for federated learning with isolated subspace training," *Advances in Neural Information Processing Systems*, vol. 36, pp. 10 876–10 896, 2023.

[61] C. Wu, S. Zhu, P. Mitra, and W. Wang, "Unlearning backdoor attacks in federated learning," in *2024 IEEE Conference on Communications and Network Security (CNS)*. IEEE, 2024, pp. 1–9.

[62] J. Xu, Z. Zhang, and R. Hu, "Detecting backdoor attacks in federated learning via direction alignment inspection," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 20 654–20 664.

[63] S. Jiang, H. Yang, Q. Xie, C. Ma, S. Wang, Z. Liu, T. Xiang, and G. Xing, "Towards compute-efficient byzantine-robust federated learning with fully homomorphic encryption," *Nature Machine Intelligence*, pp. 1–12, 2025.

[64] W. Wan, Y. Ning, Z. Huang, C. Hong, S. Hu, Z. Zhou, Y. Zhang, T. Zhu, W. Zhou, and L. Y. Zhang, "Mars: A malignity-aware backdoor defense in federated learning," *arXiv preprint arXiv:2509.20383*, 2025.

[65] W. Gill, A. Anwar, and M. A. Gulzar, "Feddefender: Backdoor attack defense in federated learning," in *Proceedings of the 1st International Workshop on Dependability and Trustworthiness of Safety-Critical Systems with Machine Learned Components*, 2023, pp. 6–9.

[66] G. Wan, Z. Shi, W. Huang, G. Zhang, D. Tao, and M. Ye, "Energy-based backdoor defense against federated graph learning," in *The Thirteenth International Conference on Learning Representations*, 2025.

[67] Z. Ma and T. Gao, “Federated learning backdoor attack detection with persistence diagram,” *Computers Security*, vol. 136, p. 103557, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167404823004674

[68] X. Wang, D. Dimitriadis, S. Koyejo, and S. Tople, “Invariant aggregator for defending against federated backdoor attacks,” 2024. [Online]. Available: https://arxiv.org/abs/2210.01834

[69] T. D. Nguyen, A. D. Nguyen, T.-H. Nguyen, K.-S. Wong, H. H. Pham, T. T. Nguyen, and P. L. Nguyen, “Fedgrad: Mitigating backdoor attacks in federated learning through local ultimate gradients inspection,” in *2023 International Joint Conference on Neural Networks (IJCNN)*, 2023, pp. 01–10.

[70] H. Yang, D. Gu, and J. He, “A robust and efficient federated learning algorithm against adaptive model poisoning attacks,” *IEEE Internet of Things Journal*, vol. 11, no. 9, pp. 16 289–16 302, 2024.

[71] C. Xu, Y. Jia, L. Zhu, C. Zhang, G. Jin, and K. Sharif, “Tdfl: Truth discovery based byzantine robust federated learning,” *IEEE Transactions on Parallel and Distributed Systems*, vol. 33, no. 12, pp. 4835–4848, 2022.

[72] T. D. Nguyen, P. Rieger, H. Chen, H. Yalame, H. Möllering, H. Fereidooni, S. Marchal, M. Miettinen, A. Mirhoseini, S. Zeitouni, F. Koushanfar, A.-R. Sadeghi, and T. Schneider, “Flame: Taming backdoors in federated learning (extended version 1),” 2023. [Online]. Available: https://arxiv.org/abs/2101.02281

[73] S. Zheng, W. Yuan, X. Wang, and L. Duan, “Adaptive federated learning via new entropy approach,” *IEEE Transactions on Mobile Computing*, vol. 23, no. 12, pp. 11 920–11 936, 2024.

[74] W. Huang, Z. Shi, M. Ye, H. Li, and B. Du, "Self-driven entropy aggregation for byzantine-robust heterogeneous federated learning," in *Forty-first International Conference on Machine Learning*, 2024.

[75] Y. Mu, "Deep leakage from gradients," *arXiv preprint arXiv:2301.02621*, 2022.

[76] B. Wang, Y. Yao, S. Shan, H. Li, B. Viswanath, H. Zheng, and B. Y. Zhao, "Neural cleanse: Identifying and mitigating backdoor attacks in neural networks," in *2019 IEEE Symposium on Security and Privacy (SP)*, 2019, pp. 707–723.

[77] M. Williams, R. Morales, K. Johnson, G. Martinez, and J. Bennett, "Entropy-based network traffic analysis for efficient ransomware detection," *Authorea Preprints*, 2024.

[78] Y. Li, M. Zhu, X. Yang, Y. Jiang, T. Wei, and S.-T. Xia, "Black-box dataset ownership verification via backdoor watermarking," *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 2318–2332, 2023.

[79] Z. Guan, M. Hu, S. Li, and A. K. Vullikanti, "Ufid: A unified framework for black-box input-level backdoor detection on diffusion models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 26, 2025, pp. 27 312–27 320.

[80] K. Zhang, G. Tao, Q. Xu, S. Cheng, S. An, Y. Liu, S. Feng, G. Shen, P.-Y. Chen, S. Ma, *et al.*, "Flip: A provable defense framework for backdoor mitigation in federated learning," *arXiv preprint arXiv:2210.12873*, 2022.

[81] Y. Gao, C. Xu, D. Wang, S. Chen, D. C. Ranasinghe, and S. Nepal, "Strip: A defence against trojan attacks on deep neural networks," in *Proceedings of the 35th annual computer security applications conference*, 2019, pp. 113–125.

[82] D. H. Nguyen, P. L. Nguyen, T. T. Nguyen, H. H. Pham, and D. A. Tran, "Fedblock: A blockchain approach to federated learning against backdoor attacks," in *2024 IEEE International Conference on Big Data (BigData)*, 2024, pp. 7981–7990.

[83] Y. Yao, L. Wang, J. Wu, J. Song, S. Chen, Z. Wang, Z. Tian, W. Chen, H. Li, and X. Li, "Fedre: A representation entanglement framework for model-heterogeneous federated learning," *arXiv preprint arXiv:2511.22265*, 2025.

[84] C. Yang and H. Zhao, "A dynamic optimization scheme based backdoor attack dpba in federated learning," in *2024 7th International Conference on Computer Information Science and Application Technology (CISAT)*, 2024, pp. 1323–1327.